

REVIEW

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A systematic review of lightweight cryptographic schemes for security and privacy in IoT

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Abstract

The rapid growth of the Internet of Things (IoT) has created significant challenges in safeguarding security and privacy in resource-constrained environments. Conventional encryption algorithms often impose heavy computational and energy costs, making them unsuitable for IoT devices. To address this, lightweight cryptographic schemes have been widely studied as efficient alternatives. This paper presents a Systematic Literature Review (SLR) following Kitchenham's guidelines, analyzing 77 peer-reviewed studies published between 2006 and 2025. Both qualitative synthesis and quantitative evaluation (e.g., entropy, execution time, energy consumption, throughput, NPCR, UACI) are conducted. The review is guided by four research questions focusing on: (i) lightweight cryptographic techniques for confidentiality, integrity, and authentication; (ii) performance trade-offs in constrained IoT devices; (iii) optimizations of Elliptic Curve Cryptography (ECC); and (iv) integration with emerging technologies such as AI, blockchain, and steganography. The findings classify lightweight block and stream ciphers, hybrid frameworks, and authentication protocols while highlighting their strengths and limitations. The study also identifies research gaps, such as limited adaptive models and insufficient benchmarking on embedded platforms. This work contributes a clear roadmap for designing secure, scalable, and energy-efficient cryptographic solutions tailored for IoT ecosystems, thereby guiding both academic research and real-world deployment.

Keywords Internet of things (IoT), Lightweight cryptography, Data security, Privacy, Secure communication, IoT layers

1 Introduction

The IoT (IoT) has rapidly evolved into a transformative paradigm, interconnecting billions of devices across domains such as healthcare, smart cities, agriculture, and industrial automation. The rapid proliferation of IoT has revolutionized domains such as healthcare, transportation, agriculture, and industrial automation by enabling real-time monitoring, automation, and intelligent decision-making. However, this expansion has also magnified the attack surface of resource-constrained devices, exposing them to increasingly sophisticated cyber threats. Conventional cryptographic solutions like



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RSA and AES-256, while offering strong security guarantees, impose significant computational, memory, and energy demands that surpass the capabilities of lightweight sensors, embedded controllers, and edge devices [1]. Public Key Infrastructure (PKI)-based mechanisms further exacerbate this issue, as they are inherently computationally intensive and energy-hungry, making them unsuitable for IoT nodes with limited memory, processing power, and battery capacity [2]. These limitations underscore the urgent need for lightweight cryptographic mechanisms that preserve confidentiality, integrity, and authentication while minimizing power consumption, latency, and resource utilization [3].

To address these challenges, researchers have proposed lightweight cryptographic solutions that ensure confidentiality, integrity, and authentication with minimal computational overhead [4]. These include optimized block and stream ciphers, elliptic curve cryptography (ECC), lightweight authentication protocols, and hybrid approaches integrating steganography, AI, and blockchain. Despite several related surveys, most either focus on computationally heavy solutions or overlook optimization-centered frameworks [5]. This paper conducts a systematic literature review (SLR) of 77 peer-reviewed studies (2006–2025), providing a taxonomy of lightweight cryptographic schemes, identifying common attack vectors, analyzing performance trade-offs, and highlighting research gaps and emerging trends. The findings aim to guide the design of robust, scalable, and resource-aware cryptographic frameworks tailored to IoT ecosystems.

Based on projections from the IoT Analytics platform, the global count of IoT-connected devices is expected to grow by 13% in 2025, reaching approximately 25.8 billion [6]. This marks a significant rise of 15.2 billion devices since 2019. Looking ahead, this figure is forecasted to double by 2030, potentially exceeding 40 billion connected devices worldwide [7].

The IoT represents a transformative technological paradigm that facilitates seamless and pervasive interaction among devices, systems, and users. By leveraging intelligent sensors, real-time data is collected from the physical environment and transmitted to specialized applications for analysis, decision-making, and automated action [8]. This integration has been widely acknowledged by modern industries as a catalyst for operational efficiency, innovation, and competitive advantage [9]. As a result, physical spaces are evolving into interconnected smart environments, driving significant advancements in automation, resource optimization, and sustainable practices [10]. The influence of IoT spans numerous domains, including smart cities, smart homes, environmental monitoring, energy optimization, intelligent transportation, education, healthcare systems, air traffic regulation, and weather prediction [11]. Furthermore, academic institutions are increasingly adopting IoT-enabled infrastructures to enhance operational sustainability, streamline campus management, and optimize daily functions through data-driven strategies [12].

IoT is witnessing exponential growth, profoundly transforming diverse sectors of contemporary life by seamlessly interconnecting intelligent systems, devices, and sensors [13]. As a transformative technological paradigm, IoT integrates previously isolated infrastructures into a unified ecosystem, thereby enhancing automation, data exchange, and real-time decision-making [14]. This integration has significantly reshaped daily routines through innovations such as smart homes, where internet-enabled devices automate tasks, optimize energy consumption, and improve residential comfort and

safety [15]. In the healthcare domain, IoT has enabled the development of Smart Health Monitoring Systems, which leverage wearable and portable biomedical devices to facilitate continuous health and fitness tracking, both in personal and clinical settings [16]. These systems support real-time patient monitoring, especially in critical care units, contributing to timely interventions and improved outcomes [17]. Additionally, IoT plays a crucial role in assistive technologies for the elderly and individuals with disabilities, offering solutions that enhance independence and quality of life [18]. The affordability and accessibility of IoT components have further accelerated their adoption across diverse socioeconomic groups [19]. In transportation, IoT has enhanced traffic management and vehicular efficiency through the deployment of intelligent traffic control systems and sensor-equipped vehicles capable of predicting congestion and suggesting optimized routes [20]. Overall, the IoT represents a foundational advancement in the digital age, unifying heterogeneous technologies to create smart, responsive, and interconnected environments across healthcare, home automation, mobility, and more [21].

The rapid evolution of wireless communication technologies such as Bluetooth, RFID, Wi-Fi, and cellular networks has catalyzed the rise of the IoT as a transformative force in the modern digital landscape [22]. Driven by the demand for seamless connectivity among billions of smart devices and services across the globe, the adoption of IoT continues to accelerate at an unprecedented pace. However, this surge in interconnectivity also brings with it a significant escalation in security challenges [23]. As IoT networks grow in scale and complexity, they become more attractive targets for adversaries equipped with increasingly advanced tools and techniques. The frequency, variety, and sophistication of cyber threats are rising steadily, exposing vulnerabilities in both device-level security and data transmission protocols [24]. Consequently, securing the IoT ecosystem is not merely an operational necessity but a foundational requirement for unlocking its full potential in critical sectors such as healthcare, agriculture, smart cities, industrial automation, and intelligent transportation systems.

In the coteremporary digital landscape, IoT has become deeply embedded in daily life, facilitating seamless connectivity across smart homes, healthcare systems, industrial automation, and transportation, agriculture, etc [25]. However, this pervasive integration also introduces significant security and privacy challenges [26]. IoT Security and privacy refers to the strategic protection of interconnected devices and their underlying infrastructure comprising sensors, embedded systems, networks, and cloud platforms against unauthorized access, data breaches, and malicious attacks [27]. The protection of sensitive user data, such as personal identifiers, financial credentials, and behavioral patterns, is critical to maintaining trust and operational reliability. Threat vectors such as malware infections, phishing attacks, ransomware, denial-of-service (DoS) incidents, and cyber-espionage increasingly target IoT ecosystems due to their heterogeneous and resource-constrained nature [28]. To mitigate these risks, security mechanisms including strong cryptographic protocols, lightweight authentication schemes, secure firmware updates, multi-factor authentication, and robust network segmentation must be deployed [29]. Furthermore, beyond prevention, IoT security frameworks must incorporate incident response and recovery mechanisms to contain damage, restore system integrity, and prevent escalation. As a result, ensuring end-to-end security and privacy in IoT is not only about defense but also about resilience, making it a cornerstone of modern digital infrastructure [30].

Cryptographic primitives are basic building blocks of security algorithms used to protect data and secure communication in IoT systems.

- Symmetric-key cryptography is a fundamental cryptographic technique in which a single shared secret key is used for both encryption and decryption processes. Its security relies on the secrecy and strength of the key, ensuring that without access to this key, it is computationally infeasible for an adversary to retrieve or alter the original plaintext from the ciphertext. This approach is particularly efficient for resource-constrained environments like IoT due to its low computational overhead.
- Public-key cryptography, also known as asymmetric cryptography, relies on a mathematically linked key pair comprising a public key and a private key. The public key is openly shared and used for encryption, while the corresponding private key, which must remain confidential, is used for decryption. This mechanism ensures that only the intended recipient who possesses the private key can access the original plaintext from the ciphertext, thereby enabling secure communication over untrusted networks and supporting critical functions like confidentiality, authentication, and digital signatures [20, 21].
- A cryptographic hash function is a one-way mathematical algorithm that maps input data of arbitrary length to a fixed-length hash value, commonly known as a message digest. To be cryptographically secure, it must satisfy three key properties: preimage resistance (infeasibility of deriving the original input from its hash), second preimage resistance (difficulty in finding a different input with the same hash), and collision resistance (impossibility of finding two distinct inputs producing identical hashes). These properties make hash functions essential for ensuring data integrity, as they enable the detection of any unauthorized or accidental alterations during storage or transmission.
- A Message Authentication Code (MAC) is a compact cryptographic checksum generated from a message and a secret key, serving to verify both the authenticity and integrity of the message. By appending the MAC to the message during transmission, the receiver can use the shared secret key to recompute and validate the MAC, ensuring that the message originated from a legitimate sender and has not been tampered with during transit.
- Finally, lightweight cryptography offers a more efficient solution tailored for low-power devices, eliminating the need for heavy certificate-based authentication. Instead of managing digital certificates to establish trust, lightweight schemes utilize compact key exchange protocols, pre-shared symmetric keys, or elliptic curve-based mechanisms with minimal computational footprint. These approaches significantly reduce latency and energy consumption while still ensuring secure communication, device authentication, and data integrity. Lightweight cryptographic solutions are more practical and scalable for large-scale IoT deployments where memory, processing power, and bandwidth are severely limited.

Five essential steps for the working of a lightweight cryptographic algorithm.

Step 1 Key Initialization.

Devices are provisioned with lightweight cryptographic keys (e.g., pre-shared symmetric keys or ECC-based public-private keys) during deployment.

Step 2 Mutual Authentication.

Upon connection, devices perform lightweight mutual authentication using hash-based challenges, MACs (Message Authentication Codes), or ECC-based handshake protocols to verify each other's identity.

Step 3 Secure Key Exchange.

A secure session key is established using an efficient method such as ECDH (Elliptic Curve Diffie-Hellman) or symmetric key derivation.

Step 4 Confidential Data Transmission.

The established session key is used with lightweight encryption algorithms to ensure confidentiality of transmitted data with minimal resource consumption.

Step 5 Integrity and Freshness Verification.

Encrypted data appended with hash and timestamps to ensure data integrity and freshness, preventing replay and tampering attacks.

1.1 Motivation and contribution

The IoT has introduced unprecedented opportunities for intelligent automation, real-time monitoring, and ubiquitous connectivity [134]. However, these advantages come at the cost of significantly expanded attack surfaces, especially since IoT nodes are often deployed in untrusted environments and operate under strict limitations of memory, energy, and computation. Conventional cryptographic mechanisms, particularly those relying on heavyweight public-key infrastructures, impose prohibitive overheads and are thus ill-suited for constrained devices. This gap underscores the urgent need for lightweight cryptographic frameworks that preserve confidentiality, integrity, and authentication while ensuring efficiency and scalability.

This survey advances the state of the art in three keyways. First, it introduces a refined classification of lightweight cryptographic solutions organized by security objectives (confidentiality, integrity, authentication, and steganography), offering a more application-oriented perspective compared to prior reviews. Second, it develops a comparative framework synthesizing insights from 77 primary studies published between 2006 and 2025, combining qualitative synthesis with quantitative performance analysis on metrics such as entropy, execution time, NPCR, UACI, throughput, and energy consumption. Third, it emphasizes emerging research directions, including the integration of lightweight cryptography with AI, blockchain, and steganography, and highlights critical gaps such as the absence of adaptive cryptographic models and post-quantum readiness. Together, these contributions provide a roadmap for researchers and practitioners to design secure, scalable, and energy-efficient cryptographic solutions tailored to dynamic IoT environments.

The rest of the paper is arranged as follows; Sect. [Related survey studies](#) presents a critical analysis of existing survey studies. Section [Research design](#) outlines the research methodology; Sect. [Results](#) synthesizes findings from primary studies. Section

Discussion answers each research question. Sect. **Finding** presents core findings. Section **Limitations of the Survey** discusses the limitations. Section **Future Direction** outlines future research directions. Finally, Sect. **Conclusion** concludes the paper.

2 Related survey studies

Recent advancements in IoT security have introduced lightweight cryptographic frameworks optimized for constrained environments by minimizing computational complexity and energy overhead. These include streamlined symmetric and asymmetric ciphers, ECC-based key exchange, and hash-driven authentication mechanisms tailored for real-time, low-power communication. Hybrid approaches integrating steganography and biometric verification further enhance confidentiality and identity assurance. Presented below are several survey studies that have been carried out in the domain of search-based testing.

Ahmad et al. [31], provides a comprehensive examination of machine learning (ML) techniques applied to various security challenges in the IoT ecosystem. The authors systematically categorize existing research based on attack types, datasets, ML algorithms, and performance evaluation metrics, highlighting the increasing reliance on supervised, unsupervised, and deep learning models for intrusion detection, anomaly detection, and malware classification in IoT networks.

Bahaa et al. [32], presents a comprehensive review of the integration of DevSecOps practices into IoT security monitoring frameworks. It highlights the challenges posed by real-time security attacks in IoT environments and investigates how DevSecOps by embedding security early and continuously in the software development lifecycle can effectively address these issues.

Yu et al. [33], offers a comprehensive analysis of existing research focused on integrating blockchain technology within IoT infrastructures for smart cities, with particular emphasis on security challenges. The authors categorize security threats into various layers, including perception, network, and application, and examine how blockchain's inherent features such as decentralization, immutability, and smart contracts can be leveraged to address these issues.

Zubaydi et al. [18], conducted a comprehensive systematic literature review to explore the integration of blockchain technology in enhancing the security and privacy of IoT systems. The paper categorizes existing research based on key security objectives confidentiality, integrity, authentication, and non-repudiation while highlighting how blockchain's decentralized and tamper-resistant architecture can address vulnerabilities inherent in resource-constrained IoT environments.

Al-Hadhrami et al. [34], presents an extensive survey of Distributed Denial-of-Service (DDoS) attacks targeting IoT environments. The authors systematically analyze existing literature, categorizing DDoS attacks based on vectors, targets, and affected network layers, while highlighting the unique vulnerabilities of IoT systems due to their constrained resources and heterogeneous architectures. The review also evaluates various detection, mitigation, and prevention techniques, providing a comparative analysis of their effectiveness, limitations, and applicability in real-world IoT scenarios.

Alkhateeb et al. [35], presents a systematic literature review focused on hybrid blockchain platforms tailored for IoT environments, aiming to address the inherent limitations of traditional blockchain systems such as latency, scalability, and resource

consumption. By analyzing a wide range of scholarly works, the study categorizes hybrid blockchain architectures those combining public and private blockchains as promising solutions to IoT-specific challenges.

Rajmohan et al. [36] presents a comprehensive survey of IoT security research conducted over the past ten years, focusing on recurring architectural patterns, design principles, and security mechanisms tailored for the unique challenges of IoT ecosystems. It categorizes and analyzes 79 peer-reviewed papers, offering insights into how patterns such as authentication, secure communication, data protection, and trust management have evolved in IoT system architectures.

Shojaei, et al. [37], presents a systematic literature review on the security and privacy challenges within health information systems (HIS). It meticulously surveys existing technologies, frameworks, and protocols implemented to protect sensitive medical data against unauthorized access, breaches, and misuse. The review categorizes security approaches into encryption, access control, blockchain, and privacy-preserving mechanisms, while highlighting the trade-offs between system performance and data protection.

Li et al. [38], presents a comprehensive systematic literature review focused on feature selection techniques used in machine learning-based attack classification for IoT security. Recognizing the resource constraints and high vulnerability of IoT environments, the authors evaluate a wide range of filter, wrapper, and embedded feature selection methods to enhance the accuracy, efficiency, and interpretability of attack detection models. The review highlights how proper feature selection not only reduces the dimensionality of datasets but also mitigates overfitting and improves detection performance.

El-Hajj et al. [39], presents a systematic literature review focused on the exploration of smooth number-based Message Authentication Code (MAC) algorithms for secure communication in IoT devices. The study analyzes the theoretical foundations and practical implementations of MACs that leverage smooth numbers integers with only small prime factors emphasizing their potential in resource-constrained environments. By reviewing 89 relevant works, the paper categorizes existing MAC schemes based on their cryptographic primitives, efficiency, resistance to attacks, and applicability in IoT scenarios.

Ding et al. [123], present a lightweight batch authentication and key agreement scheme for IIoT gateways that authenticates multiple devices simultaneously, reducing handshake overhead and latency. Using ECC with hash-based operations, the protocol ensures confidentiality, mutual authentication, and forward secrecy while resisting replay and impersonation attacks. Evaluations confirm improved computational cost, communication efficiency, and energy consumption, making it highly suitable for large-scale, resource constrained IIoT environments.

Kumar and Das [124], propose a lightweight RFID authentication protocol for edge and blockchain-enabled supply chains in IIoT. The scheme leverages mutual authentication, hash functions, and blockchain integration to ensure secure traceability, prevent replay and cloning attacks, and decentralize trust management. By offloading heavy computation to edge nodes and using optimized lightweight operations on RFID tags, the protocol significantly reduces latency, communication cost, and energy consumption, while enhancing scalability and resilience in large-scale supply chain networks.

Alabdulatif [125], introduces a blockchain-based privacy-preserving authentication and access control model for e-health systems. The scheme combines lightweight cryptographic operations with decentralized blockchain mechanisms to provide secure user authentication, fine-grained access control, and resistance against replay and impersonation attacks. By eliminating reliance on centralized authorities and incorporating privacy-preserving key management, the model reduces single points of failure and ensures scalability, integrity, and confidentiality of sensitive medical data in IoT-enabled health-care environments.

Chen et al. [126], present a lightweight authentication and key agreement protocol for IoT-enabled smart grids, designed to secure communication between smart meters and control centers. The scheme integrates elliptic curve cryptography with hash-based operations to achieve mutual authentication, session key establishment, and forward secrecy, while resisting replay, impersonation, and man-in-the-middle attacks. Performance results show reduced computational and communication overhead, making the protocol efficient and practical for large-scale, resource-constrained smart grid environments.

Khan et al. [127], propose a lightweight scalable hybrid authentication framework for the Internet of Medical Things (IoMT) that integrates blockchain hyperledger consortium networks with edge computing. The framework combines symmetric lightweight cryptography with blockchain-based decentralized trust management to provide secure, scalable, and transparent authentication among medical devices. By leveraging edge nodes for computation, it reduces latency, energy consumption, and communication overhead, while ensuring data integrity, privacy preservation, and resilience against replay and impersonation attacks in IoMT healthcare systems.

The evaluation of the current studies is to provide a comprehensive analysis of lightweight cryptographic approaches for enhancing privacy and security in IoT systems shows in Table 1.

Based on this analysis, it is evident that while existing surveys advance IoT security research through machine learning, blockchain, and domain-specific frameworks, they provide limited attention to lightweight cryptographic schemes and their optimization for constrained environments. This gap underscores the originality of our work, which systematically classifies and evaluates lightweight cryptographic primitives, explores hybrid and emerging integrations (AI, blockchain, steganography), and highlights optimization strategies that directly address the unique challenges of IoT security.

3 Research design

This paper presents a systematic literature review (SLR) on lightweight cryptographic schemes for IoT privacy and security, adhering to Kitchenham and Charters' guidelines for methodological rigor. Focusing on resource-constrained IoT environments, the study examines optimized cryptographic techniques for confidentiality, integrity, and authentication, emphasizing code coverage as a measure of vulnerability detection. Relevant studies from high-quality databases over the past two decades were analyzed using tools like Mendeley for efficient management. The review consolidates current advancements and highlights future directions for secure, efficient, and scalable lightweight cryptographic solutions in IoT.

Table 1 Comparative study of related works in IoT security and privacy

Ref.	Author(s)	Focus area	Security objective	Strengths	Limitations
[31]	Ahmad et al.	ML for IoT Security	Intrusion/Anomaly/Malware Detection	Comprehensive taxonomy of ML in IoT	Lacks focus on cryptographic methods
[32]	Bahaa et al.	DevSecOps in IoT	Real-time Threat Monitoring	Integrates security in SDLC	DevSecOps adoption is still limited
[33]	Yu et al.	Blockchain for Smart Cities	Multi-layer Security (Perception, Network, App)	Emphasizes decentralized trust	High energy and latency overheads
[18]	Zubaydi et al.	Blockchain in IoT	Confidentiality, Authentication	Targets resource-constrained IoT	General discussion, lacks optimization insight
[34]	Al-Hadhrami et al.	DDoS in IoT	Availability	Detailed DDoS taxonomy	Limited focus on mitigation in constrained devices
[35]	Alkhateeb et al.	Hybrid Blockchain for IoT	Latency, Scalability, Energy Efficiency	Tailored for IoT scalability	Trade-off between security and resource usage
[36]	Rajmohan et al.	IoT Security Architectures	System-wide Protection	10-year retrospective analysis	Architecture-level view only
[37]	Shojaei et al.	HIS Security & Privacy	Data Confidentiality in Healthcare	Focus on privacy-preserving protocols	Limited to health-care IoT
[38]	Li et al.	ML Feature Selection in IoT	Efficient Attack Detection	Enhances ML efficiency & accuracy	Focused solely on ML preprocessing
[39]	El-Hajj et al.	Smooth Number MAC for IoT	Message Authentication	Efficient for constrained devices	Still theoretical, lacks wide deployment
[123]	X. Ding et al.	IIoT-enabled WSN Key Agreement Protocol	Confidentiality, Key Agreement	Balances energy efficiency with strong security in industrial WSNs	Limited large-scale deployment validation
[124]	V. Kumar et al.	RFID-assisted Supply Chain Authentication	Authentication	Lightweight protocol ensuring secure traceability of goods	Limited resilience under high transaction loads
[125]	A. Alabdulatif	Blockchain-assisted RFID Authentication	Authentication, Integrity	Uses decentralization to prevent replay and forgery attacks	Blockchain overhead may affect ultra-low power RFID
[126]	C. Chen et al.	Cloud Storage Remote Data Authentication	Integrity, Confidentiality	Lightweight integrity verification with low communication overhead	Needs benchmarking on real cloud-IoT deployments
[127]	A. A. Khan et al.	IoT-based e-Healthcare Access Control	Access Control, Privacy	Robust, lightweight scheme safeguarding patient data under constraints	Scalability challenges in multi-user environments

3.1 Research questions

Here are three strong research questions tailored for SLR in Lightweight Cryptographic Schemes for Privacy and Security in IoT.

RQ1 What lightweight cryptographic techniques have been proposed in recent literature to ensure confidentiality, authentication, integrity, and privacy in IoT environments?

RQ2 How do lightweight cryptographic schemes compare in terms of computational complexity, energy consumption, and suitability for resource-constrained IoT devices?

RQ3 How can Elliptic Curve Cryptography (ECC) be optimized and adapted to serve as a lightweight, energy-efficient, and secure cryptographic scheme for ensuring privacy and authentication in resource-constrained IoT environments?

RQ4 What are the existing gaps and emerging trends in the integration of lightweight cryptography with steganography, AI, and blockchain for enhancing IoT security?

RQ1 focuses on identifying cryptographic primitives and protocols tailored for ensuring confidentiality, authentication, integrity, and privacy in resource-constrained IoT devices. *RQ2* evaluates these schemes in terms of computational efficiency, energy consumption, and suitability for embedded platforms, emphasizing the trade-off between security strength and resource optimization. *RQ3* investigates how Elliptic Curve Cryptography (ECC) can be optimized and applied as a lightweight cryptographic solution for securing resource-constrained IoT environments. *RQ4* investigates the integration of lightweight cryptography with emerging technologies like AI, blockchain, and steganography to enhance adaptive security, intelligent key management, and data concealment in scalable and heterogeneous IoT systems.

In the domain of IoT security, this study emphasizes the deployment of lightweight cryptographic schemes tailored for resource-constrained environments. These schemes serve as a strategic defense mechanism to counteract and minimize the impact of diverse cryptographic attacks. Through comprehensive cryptographic evaluation, the proposed approach ensures secure data transmission and privacy preservation across interconnected IoT devices while maintaining computational efficiency.

3.2 Quality assessment for search string

The reliability of the keyword selection used to formulate the search string is validated through a carefully curated set of primary research articles. This set was constructed via a systematic review process incorporating an ad-hoc search strategy. Specifically, fifteen influential studies papers focusing on lightweight cryptographic schemes for IoT security were identified and included (see Table 2). These studies were chosen from the top-ranked search results and were ensured to originate from reputable journals with substantial citation counts. Moreover, the selection encompasses a diverse range of lightweight cryptographic algorithms to maintain comprehensive coverage. To ensure the keyword string effectively retrieves all relevant IoT security papers involving various lightweight encryption techniques, the final keyword set integrates terminology derived directly from the selected high-impact articles.

3.3 Search process and selection of data source

- (i) Identifying synonyms and related terms for the keywords used in the research questions, thereby reducing the impact of variations in terminology.
- (ii) Synonyms and alternative terms are combined using the Boolean OR operator.
- (iii) Key terms are linked together using the Boolean AND operator.

Table 2 Summary of studies identified lightweight cryptographic schemes for IoT security for keywords accuracy

Ref.	Title	Journal/source
[40]	Lightweight Cryptography in IoT Networks: A Survey	Future Generation Computer Systems (ScienceDirect)
[41]	Cryptography Algorithms for Enhancing IoT Security	IoT and Cyber-Physical Systems (ScienceDirect)
[42]	Comparative performance analysis of lightweight cryptography algorithms for IoT sensor nodes	IEEE IoT Journal
[43]	Symmetrical Cryptographic Algorithms in the Lightweight IoT	International Journal of Information and Communication Technology
[44]	A Survey of Lightweight Cryptographic Algorithms for IoT-Based Applications	Springer
[45]	A Lightweight Cryptography Algorithm for Secure Smart Cities and IoT	Electrical Journal
[46]	Lightweight Cryptography for IoT: A Review	EAI Endorsed Transactions on IoT
[47]	Lightweight Cryptography Algorithms for Security of IoT Devices: A Survey	International Research Journal of Engineering and Technology
[48]	A Systematic Technical Survey of Lightweight Cryptography on IoT Environment	International Journal of Scientific & Technology Research
[49]	New Ultra-Lightweight IoT Encryption Algorithm Using Novel Chaotic System	International Journal on Technical and Physical Problems of Engineering
[50]	Enhancing IoT security with a DNA-based lightweight cryptography system	Scientific Reports
[51]	Hash Functions and Benchmarks for Resource Constrained Passive Devices: A Preliminary Study	IEEE Xplore

Population IoT, Internet of Things, smart devices, constrained devices, embedded systems, sensor nodes, and wireless sensor networks, medical IoT, industrial IoT, cyber-physical systems.

Intervention lightweight cryptography, lightweight encryption, symmetric encryption, asymmetric encryption, stream cipher, block cipher, lightweight key exchange, lightweight authentication, energy-efficient encryption, hardware-friendly cryptography, ultra-lightweight algorithms.

Outcome security, privacy, confidentiality, data integrity, authentication, resistance to attacks, reduced memory usage, reduced energy consumption, increased throughput, enhanced protection, secure communication.

We executed our search process in two distinct stages, as illustrated in Fig. 1. Initially, we performed an electronic database search, followed by a manual search of selected conference proceedings and reputed journals to ensure comprehensive coverage. The detailed results derived from electronic databases are presented in Table 2.

To comprehensively incorporate the most influential and relevant sources into our systematic literature review (SLR) (Fig. 1), an initial manual exploration was conducted across selected high-impact journals and conference proceedings [52]. These publication venues were specifically targeted due to their established history of disseminating pivotal primary research closely aligned with our domain. Notably, the chosen sources include journals such as international journal of Information technology, IEEE Transactions on Arabian journal of science and engineering, cluster computing, ACM Transactions on Software Engineering and Methodology the Journal of Ambient Intelligence and Humanized Computing, and Soft Computing. Additionally, significant conference



Fig. 1 Two stages search process

proceedings such as the International Conference on Computing for Sustainable Global Development (INDIACom), International Conference on Artificial Intelligence and Emerging Technology (Global AI Summit), Cybersecurity for industrial IoT: architecture, models and lessons learned, IEEE World Forum on IoT were systematically reviewed due to their prominence and consistent publication of relevant studies within the research landscape.

3.4 Study selection

To achieve comprehensive and precise coverage in our systematic survey, the search strings were iteratively refined during the initial search phase to eliminate irrelevant results effectively. This iterative optimization notably enhanced the relevance and quality of the retrieved literature, effectively addressing the complex task of balancing thoroughness with precision. Articles published between 2006 and April 2025 were considered, applying well-defined inclusion criteria across our search strategy. Different variations of search strings were utilized to target titles, abstracts, and full-text content distinctly. In contrast, custom search strings were formulated separately for databases like SpringerLink due to their unique search interfaces. The complete set of finalized search strings is detailed in Table 4, highlighting key terms specifically relevant to lightweight cryptography, such as "security," "privacy," "confidentiality," "authentication," "integrity," "encryption," "lightweight," "IoT," "resource-constrained," "cryptanalysis," "performance," "efficiency," "key management," "attack resistance," and "algorithm."

The structured search string formulated specifically for this study is provided as follows:

("IoT" OR "Internet of Things" OR "embedded systems") AND ("lightweight cryptography" OR "lightweight encryption" OR "lightweight security protocols") AND ("security" OR "privacy" OR "confidentiality" OR "authentication" OR "data integrity" OR "efficiency" OR "low-power" OR "steganography").

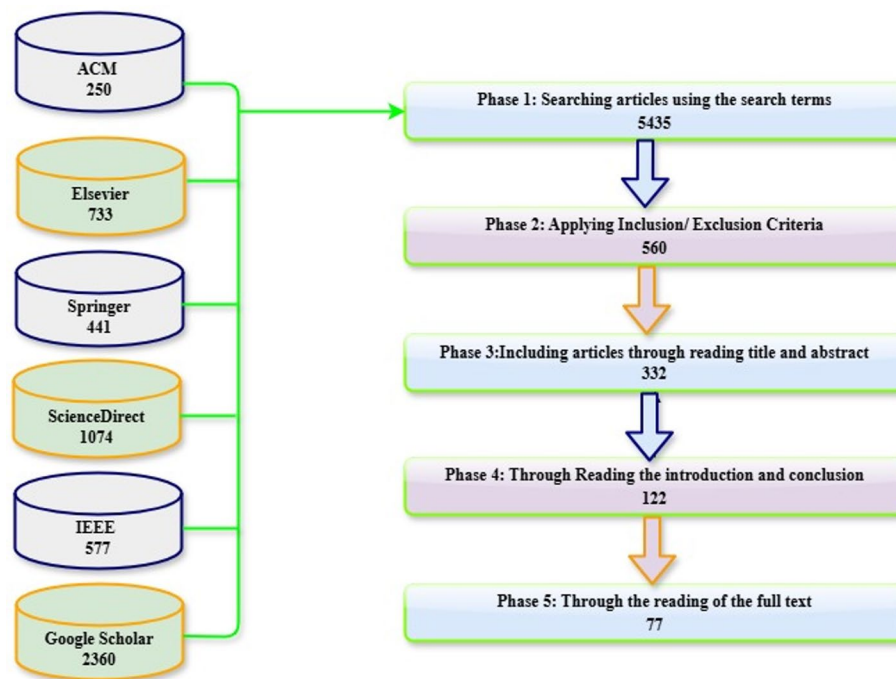


Fig. 2 Steps Involved in Search Strategy via multiple filtration

Table 4 The compilation of databases chosen for the systematic literature review and the results obtained post-searching via the search string

Databases	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5 (Final Selection)	Percentage (%)
Google scholar	2360	180	110	40	24	31.17
IEEE Xplore digital library	577	100	75	30	19	24.68
Sciencedirect	1074	120	60	20	11	14.29
Springer nature	441	60	35	15	8	10.39
Elsevier	733	65	30	10	7	9.09
ACM digital library	250	35	22	7	8	10.39
Total	5435	560	332	122	77	100%

To ensure methodological rigor and relevance, we established a set of clearly defined inclusion criteria aligned with the systematic review process proposed by Kitchenham and Charters in Fig. 2. The studies were selected based on the following parameters:

Inclusion Criteria 1 The study must be published in the English language, ensuring accessibility and consistency in interpretation.

Inclusion Criteria 2 The publication must fall within the timeframe of 2006 to 2025, encompassing both foundational research and contemporary developments in lightweight cryptography for IoT security.

Inclusion Criteria 3 The study must propose a lightweight cryptographic procedure, methodology, framework, tool, relevant to security and privacy enhancement in IoT systems.

Inclusion Criteria 4 It discusses the proof of security to mitigate the various kind of attacks.

Inclusion Criteria 5 The study must address at least one specific security attributes such as confidentiality, authentication, integrity, steganography within the context of IoT systems.

Inclusion Criteria 6 The study must describe the security tools to address the lightweight cryptography.

Articles meeting any of the following criteria will be excluded from the study:

Exclusion Criteria 1 Studies not published in English.

Exclusion Criteria 2 Articles that only propose theoretical guidelines without implementing lightweight cryptographic schemes.

Exclusion Criteria 3 Studies that are not directly related to lightweight cryptography, IoT security, or do not address the defined research questions.

Exclusion Criteria 4 Grey literature such as opinion pieces, viewpoints, editorials, prefaces, discussions, tutorials, and presentations.

Exclusion Criteria 5 Duplicate entries (in cases of multiple versions, only the latest and most comprehensive version will be retained).

Exclusion Criteria 6 Articles where lightweight cryptographic schemes or IoT security are not the core focus but are only mentioned briefly as examples.

Exclusion Criteria 7 Book chapters focused on unrelated aspects of security rather than lightweight cryptographic methods.

Exclusion Criteria 8 Survey papers that provide overviews of security without experimental validation of lightweight cryptographic solutions in IoT.

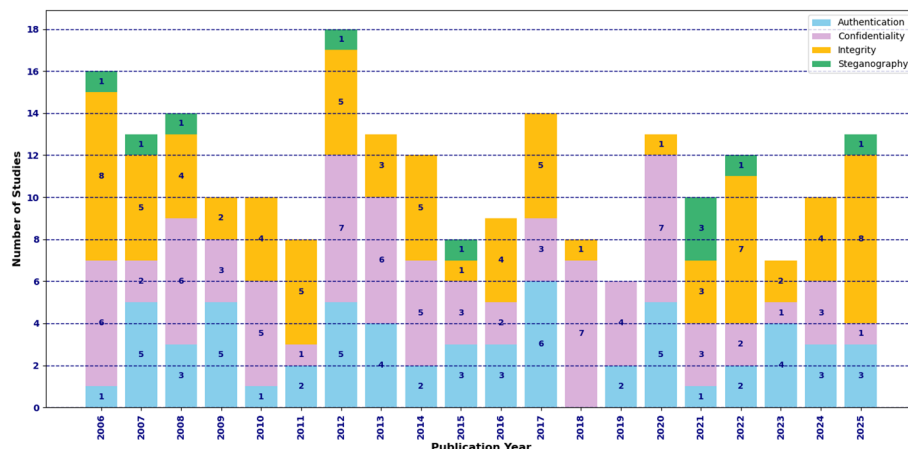
3.5 Study filtering

We adopted a structured search strategy based on predefined keywords relevant to lightweight cryptographic schemes for privacy and security in IoT environments (Table 2). As shown in Fig. 2, Phase 1 involved querying multiple scholarly databases, including Google Scholar (2360), ScienceDirect (1074), Elsevier (733), IEEE (577), Springer (441), and ACM (250), resulting in a total of 5435 articles. In Phase 2, the inclusion and exclusion criteria were applied, filtering the records down to 560 articles by eliminating non-English studies, duplicate entries, irrelevant topics, and grey literature. Next, in Phase 3, through title and abstract screening, 332 articles were selected that closely aligned with the review objectives. In Phase 4, the introduction and conclusion sections of these papers were thoroughly reviewed, leading to the shortlisting of 122 articles. Finally, in Phase 5, after a complete reading of the full texts and applying quality assessment procedures, 77 primary research articles were identified for inclusion in this systematic review (Table 3). These selected papers formed the final dataset for evaluating the effectiveness, applicability, and limitations of lightweight cryptographic techniques in IoT-based security frameworks.

Crafting an effective and exhaustive search string to retrieve all primary studies on lightweight cryptographic schemes for IoT security remains a complex task, primarily due to inconsistent terminology and varied author expressions across publications. This limitation became evident in Phase 5 of our search strategy (Fig. 3), where additional papers were identified through reference scanning of previously shortlisted studies. For example, several works applied optimization techniques such as genetic algorithms within lightweight encryption frameworks but lacked explicit references to “security” or

Table 3 Search string for different databases

Databases	Search string
Google Scholar	"IoT security" AND "lightweight cryptography"
IEEE Xplore Digital Library	((("IoT" OR "Internet of Things" OR "embedded systems") AND ("lightweight cryptography" OR "lightweight encryption" OR "lightweight security protocols") AND ("security" OR "privacy" OR "confidentiality" OR "authentication" OR "data integrity" OR "efficiency" OR "low-power" OR "steganography"))
ScienceDirect	("IoT" OR "Internet of Things") AND ("lightweight cryptography" OR "lightweight encryption") AND ("security" OR "privacy" OR "authentication")
Springer Nature	("lightweight cryptography" AND "Internet of Things") AND ("privacy" OR "data integrity")
Elsevier	((("IoT" OR "Internet of Things") AND ("lightweight cryptography") AND ("privacy" OR "authentication" OR "integrity"))
ACM digital Library	("Internet of Things" OR IoT OR "embedded systems") AND ("lightweight cryptography" OR "lightweight encryption" OR "lightweight security protocol") AND ("privacy" OR "data security" OR "confidentiality" OR "authentication" OR "data integrity")

**Fig. 3** Year-wise distribution of security measure in lightweight cryptographic research in IoT

"lightweight cryptography" in their titles, abstracts, or keywords. Such omissions would have caused these relevant studies to be excluded if only standard database search strings were used. This emphasizes the critical role of manual reference scanning in uncovering pertinent literature that may otherwise be missed due to vague or inconsistent indexing. Consequently, we meticulously reviewed every paper during this phase to ensure comprehensive coverage of studies directly contributing to lightweight cryptographic schemes for privacy and security in IoT.

3.6 Criteria for quality assessment

In this systematic review of lightweight cryptographic schemes for privacy and security in IoT, the quality assessment considered both quantitative and qualitative studies, as no restrictions were imposed on specific research methods or experimental designs. Ensuring the inclusion of high-quality studies was essential to support accurate data analysis, meaningful results, and informed future research. To reduce the risk of misinterpretation due to low-quality studies, a structured evaluation process was conducted, as outlined in Table 5. Only those studies that offered thorough experimental evidence and detailed evaluations were selected. A set of targeted assessment questions was developed

Table 5 Quality assessment

S. No.	Evolution score
QA1	Yes = 1, No = 0, Partial = 0.5
QA1	Yes = 1, No = 0, Partial = 0.5
QA1	Yes = 1, No = 0, Partial = 0.5
QA1	Yes = 1, No = 0, Partial = 0.5

to examine the methodological strength, clarity of results, and relevance to lightweight cryptographic approaches in IoT. Each study was reviewed based on these criteria.

QA1 Does the study propose or evaluate lightweight cryptographic schemes aimed at enhancing IoT security or privacy?

QA2 Does the study define and explain the evaluation metrics or performance criteria (e.g., entropy, execution time, energy consumption) used to assess the effectiveness of the cryptographic scheme?

QA3 Are the algorithmic details and design methodology of the proposed lightweight cryptographic scheme clearly and systematically described?

QA4 Has the proposed scheme been experimentally validated using real-world IoT data, simulations, hardware platforms (e.g., Raspberry Pi, Arduino), or benchmark datasets?

To ensure methodological rigor, each primary study was evaluated using four quality assessment criteria scored as Yes (1), Partial (0.5), and No (0), with a maximum of 4. Primary studies scoring ≥ 3.0 were selected for inclusion. This scoring framework ensured that only technically sound and empirically validated primary studies were included in the final review.

3.7 Collection of data

From each selected primary study, the following critical information was systematically extracted to ensure a comprehensive review of lightweight cryptographic solutions in IoT contexts:

- The names of the authors were noted to identify the origin and affiliation of the research.
- The publication venue, whether a journal or conference proceeding, was recorded to understand the scholarly context and credibility of the work.
- The article's year of publication was captured to establish the research's relevance and timeline within the evolving domain of IoT security.
- The paper's references were examined to assess the depth of literature grounding and citation network.
- The specific security concern addressed—such as confidentiality, integrity, authentication, or performance criteria like throughput or energy efficiency—was identified.
- If any cryptographic design methodology involved optimization strategies (e.g., genetic algorithms, firefly, PSO), these were documented.
- The performance evaluation criteria (e.g., entropy, PSNR, UACI, execution time, and memory usage) used to assess the effectiveness of the lightweight cryptographic scheme were reviewed.

- The main objective of the study and its formulated research questions were noted to understand the research scope.
- Detailed technical information about the proposed cryptographic method was extracted, including any algorithmic or architectural innovations.
- Whether the proposed method was experimentally evaluated or benchmarked against existing cryptographic techniques was noted.
- The key findings and conclusions were summarized, along with insights into directions for future work.
- Any stated limitations, constraints, or challenges in implementing the proposed approach were also recorded to understand the boundaries of the study's applicability.

3.8 Analysis of data

The extracted data from the selected studies were systematically organized and analyzed to address the defined research questions of this review:

- For RQ1, the analysis identifies a range of lightweight cryptographic techniques proposed in recent literature that focus on achieving key security objectives such as confidentiality, authentication, integrity, and privacy within IoT environments.
- In response to RQ2, the selected studies were compared based on their computational complexity, energy consumption, and overall performance to determine their suitability for deployment in resource-constrained IoT devices.
- To address RQ3, particular attention was given to how Elliptic Curve Cryptography (ECC) has been optimized and adapted across various implementations to act as a lightweight, secure, and energy-efficient solution for privacy-preserving communication and authentication in IoT environments.
- Regarding RQ4, the review also explores current research gaps and highlights emerging trends that involve the integration of lightweight cryptography with steganography, artificial intelligence techniques, and blockchain technologies to enhance end-to-end IoT security frameworks.

4 Results

This section addresses the defined research questions by synthesizing insights from the final set of selected studies. A total of 77 primary research articles were analyzed to understand the landscape of lightweight cryptographic solutions in IoT. These studies propose and evaluate cryptographic techniques aimed at ensuring confidentiality, integrity, authentication, steganography. The analysis reveals that various cryptographic methods such as stream ciphers, block ciphers, hash-based schemes, and ECC-based protocols have been applied and assessed for their performance on resource-constrained IoT devices. Among them, several studies focus specifically on optimizing Elliptic Curve Cryptography to reduce computational overhead and enhance energy efficiency while maintaining strong security properties. In addition, recent efforts have explored hybrid approaches by integrating lightweight cryptography with steganography, artificial intelligence, and blockchain to address evolving IoT security challenges and uncover potential areas for future research. The distribution and categorization of these studies are illustrated in the corresponding Figs. 3 and 4.

In this systematic review of lightweight cryptographic schemes for IoT security, analysis of 77 primary research articles shows varied focus on key security objectives.

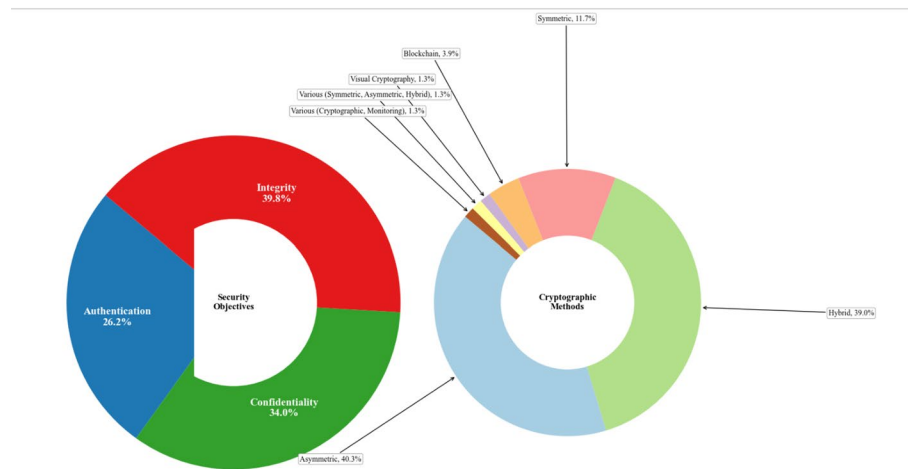


Fig. 4 Proportional representation of security measure in lightweight cryptographic studies in IoT

Authentication is addressed in 50 studies, reflecting its importance in verifying device identity and ensuring secure communication. Confidentiality covered by 65 and integrity are each covered in 76 studies, underscoring their fundamental role in protecting data from unauthorized access and tampering. In comparison, steganography appears in only 8 studies, indicating it is a less commonly adopted approach, primarily used for data concealment rather than core cryptographic protection. This highlights a clear preference in current research toward encryption and authentication over data hiding techniques.

4.1 Selected primary studies

In this systematic survey of 77 selected research papers focusing on lightweight cryptographic solutions for IoT security and privacy, a detailed classification of publication venues was performed to understand the distribution of scholarly contributions in Table 6. Out of the total studies, 70 papers were published in reputed journals such as IEEE Access, Multimedia Tools and Applications, Sensors, and Computers and Electrical Engineering, reflecting a strong inclination toward peer-reviewed archival research. The remaining 7 papers appeared in conference proceedings and workshops like INDIA-Com, ICSPN, and EAI International Conference, showcasing ongoing discussions and emerging innovations in cryptographic techniques. This distribution highlights the growing academic interest in secure and efficient IoT communication, with journals serving as a primary medium for disseminating validated research findings, while conferences contribute to the exploration of novel ideas and real-time discussions within the scientific community.

4.2 Authentication

Authentication emerges as a technically critical component to ensure the legitimacy of devices, users, and data sources in highly decentralized and resource-constrained IoT ecosystems. The survey identifies that 60 out of 77 studies implement authentication as a foundational security service. Technically, these solutions leverage a combination of asymmetric cryptography, elliptic curve cryptography (ECC), biometric verification, blockchain-based trust models, and hybrid protocols. ECC is particularly dominant due to its ability to provide equivalent security with shorter key lengths, thereby reducing

Table 6 Summary of journals and conferences for primary studies

Journal	Conference	Number of papers
IEEE access	-	8
International journal of information technology	-	4
Multimedia tools and applications	-	4
Journal of information security and applications	-	3
Sensors	-	3
International journal of experimental research and reviews	-	2
Concurrency and computation: practice and experience	-	2
ETRI journal	-	2
Computers and electrical engineering	-	2
International journal of communication systems	-	2
IEEE transactions	-	2
Chinese journal of electronics	-	2
Computer standards & interfaces	-	2
Pervasive and mobile computing	-	1
Computer systems science & engineering	-	1
Energies	-	1
Scientific programming	-	1
The journal of supercomputing	-	1
Biomedical signal processing and control	-	1
IoT	-	1
Applied sciences	-	1
Arabian journal for science and engineering	-	1
Wireless communications and mobile computing	-	1
IEEE transactions on wireless communications	-	1
IEEE access	-	1
High-confidence computing	-	1
IEEE access	-	1
Neural computing and applications	-	1
Soft computing	-	1
Future internet	-	1
Computers, materials & continua	-	1
Computers in human behavior	-	1
IEEE transactions on industrial informatics	-	1
Information sciences	-	1
Future generation computer systems	-	1
Journal of theoretical and applied information technology	-	1
Cluster computing	-	1
International journal of electrical and computer engineering research	-	1
Journal of King Saud university-computer and information sciences	-	1
Expert systems with applications	-	1
ACM transactions on internet technology	-	1
IEEE systems journal	-	1
Optics & laser technology	-	1
Information processing & management	-	1
-	Cyber security, privacy and networking: proceedings of ICSPN	1
-	International conference on advanced network technologies and intelligent	1

Table 6 (continued)

Journal	Conference	Number of papers
-	International conference on computing for sustainable global development (INDIACom)	1
-	International symposium ELMAR	1
-	International conference on artificial intelligence and emerging technology	1
-	International conference on advanced network technologies and intelligent computing	1
-	EAI international conference	1

computational and memory overhead, which is essential for constrained devices. Multi-factor authentication schemes using biometrics are also highlighted, especially in health-care applications, where identity assurance is critical.

Several protocols utilize lightweight mutual authentication schemes where both the IoT device and the server verify each other's legitimacy using nonce-based challenges, hash functions, or ECC-based key agreement techniques. Authentication schemes often incorporate session key generation, relying on ephemeral values to ensure forward secrecy and prevent replay attacks. The use of fuzzy extractors, attribute-based encryption, and swarm optimization techniques for authentication key generation and management shows the integration of AI and adaptive systems into cryptographic authentication.

Additionally, authentication is not treated as an isolated function, but is tightly coupled with confidentiality and integrity, forming a triad to secure end-to-end IoT communication. This coupling is evident in hybrid protocols where authentication also initiates secure key exchange and access control enforcement. In high-risk applications like medical sensor networks and industrial IoT, robust authentication mechanisms are further augmented with blockchain and generative AI to decentralize trust verification and improve scalability.

Table 7 presents a comparative overview of 15 key studies on lightweight authentication protocols in IoT, highlighting the authors, publication years, algorithms used, and core mathematical constructs. Ali and Anwer [56] proposed a hybrid AES–ECC framework using MAC and encrypted session keys, while Patel et al. [59] incorporated fuzzy logic for decision-based authentication. Ostad-Sharif et al. [62] and Chen and Chen [87] utilized ECC with anonymous identity and biometric binding, respectively. Ehuil et al. [63] employed visual cryptography, and Bao and You [69] introduced a blockchain-fuzzy extractor model. Singh et al. [77] applied ECC in drone, IoT, and cloud scenarios with nonce and signature-based mechanisms. Zulkifl et al. [58] integrated fuzzy logic with blockchain, and Rostampour et al. [80] presented ECCbAP for edge devices. Chen Y and Chen J [87] focused on biometric key generation, Luqman and Faridi [54] enhanced Diffie–Hellman for fog environments, Yang et al. [105] used ECC with trusted tokens. Collectively, these studies demonstrate the evolution of authentication schemes balancing lightweight computation with cryptographic robustness.

The comprehensive evaluation of lightweight authentication protocols presented in Table 7 highlights the progressive shift toward optimizing security in

Table 7 Summary of lightweight authentication protocols in IoT security

Ref.	Algorithm used	Mathematical description	Application	Limitation
[53]	ChaosForge	Chaotic stream cipher using key-controlled logistic maps	IoT Communication	Limited evaluation in health-care contexts
[54]	Diffie-Hellman	DH: $g^a \bmod p$; SharedKey = $g^{ab} \bmod p$	Fog-assisted IoT	Complexity in fog environments
[55]	RSA Variants	RSA: $C = M^e \bmod n$; $M = C^d \bmod n$	General IoT	Resource-intensive for IoT devices
[56]	AES	AES: Keyed block cipher with S-box substitution and MixColumns	Healthcare IoT	Limited scalability for large networks
[57]	IOTA Tangle		Environmental IoT	Scalability issues with IOTA
[58]	Fuzzy Logic		Healthcare IoT	Computational complexity
[59]	Fuzzy Logic		Smart Healthcare	Complexity in fuzzy systems
[60]	AES	AES: Keyed block cipher with S-box substitution and MixColumns	Telehealth	Lightweight but limited scalability
[61]	Fully Homomorphic Encryption	Mathematical description not available.	Medical Record Sharing	High computational overhead
[62]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Telemedicine	Limited to TMIS environments
[63]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Telemedicine	Biometric data privacy concerns
[64]	Chaotic Map		Telemedicine	Limited scalability
[65]	El-Gamal		Telemedicine	RFID-specific limitations
[66]	Chaotic Map		Telemedicine	Resource-intensive chaotic maps
[63]	Visual Cryptography		IoT-Cloud	Limited to image-based data
[67]	RSA	RSA: $C = M^e \bmod n$; $M = C^d \bmod n$	General IoT	Computational overhead of RSA
[68]	XRSA	RSA: $C = M^e \bmod n$; $M = C^d \bmod n$	General IoT	High computational cost for resource-constrained devices
[69]	Fuzzy Extractor		IoT Authentication	Complexity of fuzzy extractors
[69]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	General IoT	ECC more efficient but RSA widely used
[70]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Mobile IoT	Optimization complexity
[71]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	General IoT	Energy consumption varies by curve
[72]	AES	AES: Keyed block cipher with S-box substitution and MixColumns	General IoT	Limited to lightweight cryptography
[73]	ABE		Healthcare IoT	Computational complexity of ABE
[74]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Internet of Drones	Drone-specific constraints
[75]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Internet of Drones	Limited to FANET
[76]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	General IoT	Experimental nature of DNA cryptography
[77]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	IoT and Cloud	Limited scalability
[78]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	General IoT	Experimental DNA approach

Table 7 (continued)

Ref.	Algorithm used	Mathematical description	Application	Limitation
[56]	AES	AES: Keyed block cipher with S-box substitution and MixColumns	General IoT	Limited scalability
[79]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	IoT Security	Malware detection complexity
[80]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	IoT Edge Devices	Limited to edge devices
[81]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	General IoT	Complexity of logistic regression
[82]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Industrial IoT	Limited to industrial contexts
[16]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Real-time IoT	Hardware-specific limitations
[69]	Genetic Algorithm	Mathematical description not available.	IoT Applications	Optimization complexity
[91]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Medical Sensor Data	Limited to sensor data
[92]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Smart Homes	Limited to smart home applications
[93]	Lightweight Cryptography	Mathematical description not available.	Environmental Monitoring	Limited scalability for large-scale IoT networks
[72]	Homomorphic ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	IoT Data Aggregation	Computational overhead due to privacy-preserving mechanisms
[72]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Military IoD	Potential vulnerability to drone impersonation attacks
[83]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	General IoT	Limited scalability for large-scale IoT networks
[84]	XRSA	RSA: $C = M^e \bmod n$; $M = C^d \bmod n$	IoT Sensor Networks	Potential computational overhead for resource-constrained devices
[85]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	General IoT	Experimental nature of DNA-based cryptography
[63]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	IoT-Cloud	Limited scalability for resource-constrained IoT devices
[86]	AKA	Mathematical description not available.	General IoT	Limited focus on specific algorithm performance
[87]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Healthcare IoT	Privacy concerns with biometric data
[62]	ECC	ECC: $P = xG$; Key = $H(Px \parallel \text{Nonce})$; Auth = $\text{Sign}(P \parallel N1 \parallel N2)$	Telemedicine	Limited scalability for large-scale TMIS deployments
[88]	Hash, XOR		IoT-Cloud	Vulnerable to insider and impersonation attacks
[89]	EC-ElGamal cryptosystem		IoT Data Integrity	High computational and storage overhead
[90]	Deep Learning		Industrial IoT	Complexity of integrating security across diverse IIoT systems

resource-constrained IoT environments. Among the surveyed methods, Elliptic Curve Cryptography (ECC) stands out as the most frequently employed due to its efficiency in generating compact keys and providing robust authentication through scalar multiplication ($P = xG$) and hash-based key derivation. It has been widely adopted in applications such as telemedicine, IoT-cloud systems [128], and drone-based communications, where

emphasizing the importance of choosing optimized lightweight solutions for secure and scalable IoT deployments.

4.3 Confidentiality

In the context of lightweight cryptographic schemes for IoT security, confidentiality is the most consistently addressed security objective, with all 77 reviewed studies incorporating it as a fundamental component. Confidentiality ensures that sensitive data transmitted across IoT networks remains inaccessible to unauthorized entities, particularly in environments with highly constrained devices such as sensors, wearables, and embedded systems. To achieve this, researchers have employed a variety of lightweight encryption algorithms including AES variants, elliptic curve cryptography (ECC), tailored for minimal computational and energy overhead. The survey reveals that many studies also explore hybrid approaches combining symmetric and asymmetric encryption or integrate advanced techniques like chaotic maps, DNA encoding, and homomorphic encryption to strengthen confidentiality while preserving resource efficiency. Several models introduce data partitioning, selective encryption, and key segmentation strategies to reduce redundancy and optimize cipher performance. Moreover, confidentiality is often enforced at multiple levels, such as in end-to-end communication, edge data processing, and secure cloud offloading. The pervasive emphasis on confidentiality across diverse application domains ranging from telemedicine and industrial IoT to smart homes and environmental monitoring underscores its criticality in safeguarding private data against eavesdropping, inference attacks, and unauthorized disclosures in heterogeneous and decentralized IoT.

Table 8 represent the most impactful contributions to confidentiality in lightweight cryptographic schemes for IoT. Imam et al. [67, 68] laid foundational work using RSA and its variants for general IoT environments, while Bokhari et al. [53] proposed ChaosForge, a stream cipher leveraging chaotic dynamics for secure IoT communication. Luqman and Faridi [54] addressed fog-assisted networks with a Diffie–Hellman-based hybrid model. Ali and Anwer [56] focused on AES-driven confidentiality in healthcare IoT, and Gangwani et al. [57] integrated IOTA blockchain for secure environmental sensing. Faragallah et al. (2021) applied ChaCha20 in telehealth, and Zulkifl et al. [58] and Patel et al. [59] introduced fuzzy logic-based secure frameworks. Wenhua et al. [91] proposed a lightweight AES-based model for telehealth, while Reddi et al. [61] utilized fully homomorphic encryption for medical data confidentiality. Ostad-Sharif et al. [62] and Chen and Chen (2023) developed ECC-based TMIS protocols safeguarding data privacy. Suresh Kumar et al. [119] integrated chaotic maps for confidentiality in telemedicine. These works reflect strong methodological rigor and practical relevance, establishing confidentiality as a core requirement across various IoT domains such as healthcare, fog computing, and smart environments.

Table 8 provides a comprehensive analysis of confidentiality-oriented lightweight cryptographic schemes for diverse IoT applications. Traditional algorithms such as RSA and AES remain foundational but exhibit significant limitations including high computational cost, poor scalability, and latency, rendering them less suitable for constrained IoT environments. Variants like XRSA and Homomorphic Encryption aim to enhance security but suffer from overhead and complexity, especially in real-time or embedded

Table 8 Summary of confidentiality in lightweight cryptographic schemes for IoT security

Ref.	Algorithm used	Application	Mathematical description	Limitation
[67]	RSA	General IoT	RSA encryption uses modular exponentiation: $C = M^e \bmod n$	Computational overhead of RSA
[68]	XRSA	General IoT	RSA encryption uses modular exponentiation: $C = M^e \bmod n$	High computational cost for resource-constrained devices
[53]	ChaosForge	IoT Communication	ChaosForge employs chaotic sequences: $x_{n+1} = r * x_n * (1 - x_n)$	Limited evaluation in healthcare contexts
[54]	Diffie-Hellman	Fog-assisted IoT	DH key exchange: $K = (g^{ab}) \bmod p$ ensures shared secret	Complexity in fog environments
[68]	RSA Variants	General IoT	RSA encryption uses modular exponentiation: $C = M^e \bmod n$	Resource-intensive for IoT devices
[56]	AES	Healthcare IoT	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	Limited scalability for large networks
[57]	IOTA Tangle	Environmental IoT	IOTA uses DAG-based structure with MAM masking for secure messaging	Scalability issues with IOTA
[58]	ChaCha20	Telehealth	ChaCha20: keystream derived using quarter-round function on state matrix	Limited to video data
[58]	Fuzzy Logic	Healthcare IoT	Fuzzy systems compute decision boundaries using membership functions	Computational complexity
[59]	Fuzzy Logic	Smart Healthcare	Fuzzy systems compute decision boundaries using membership functions	Complexity in fuzzy systems
[91]	AES	Telehealth	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	Lightweight but limited scalability
[61]	Fully Homomorphic Encryption	Medical Record Sharing	FHE allows $E(f(x)) = f(E(x))$ using lattice-based encryption	High computational overhead
[62]	ECC	Telemedicine	ECC: $C = M + kP$, decryption uses d and base point P	Limited to TMIS environments
[92]	ECC	Image Security	ECC: $C = M + kP$, decryption uses d and base point P	High computational cost for images
[93]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	ECC more efficient but RSA widely used
[94]	ECC	Mobile IoT	ECC: $C = M + kP$, decryption uses d and base point P	Optimization complexity
[64]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	Energy consumption varies by curve
[72]	AES	General IoT	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	Limited to lightweight cryptography
[68]	ABE	Healthcare IoT	Mathematical description not available.	Computational complexity of ABE
[85]	DNA-based	Cloud IoT	Mathematical description not available.	Experimental stage of DNA cryptography
[95]	Chaotic Map	Cloud Storage	Chaotic map uses logistic function: $x_{n+1} = r * x_n * (1 - x_n)$	Limited to image data
[96]	Skew Tent Map	General IoT	Mathematical description not available.	Limited application scope
[97]	PSO	Medical Data Security	Mathematical description not available.	Optimization complexity
[74]	ECC	Internet of Drones	ECC: $C = M + kP$, decryption uses d and base point P	Drone-specific constraints
[75]	ECC	Internet of Drones	ECC: $C = M + kP$, decryption uses d and base point P	Limited to FANET
[98]	Chaotic Map	Image Security	Chaotic map uses logistic function: $x_{n+1} = r * x_n * (1 - x_n)$	Limited to image data
[99]	Cellular Automata	Data Hiding	Mathematical description not available.	Limited to dual images

Table 8 (continued)

Ref.	Algorithm used	Application	Mathematical description	Limitation
[64]	ECC	Image Security	ECC: $C = M + kP$, decryption uses d and base point P	High computational cost
[76]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	Experimental nature of DNA cryptography
[77]	ECC	IoT and Cloud	ECC: $C = M + kP$, decryption uses d and base point P	Limited scalability
[78]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	Experimental DNA approach
[100]	Genetic Algorithm	Cloud IoT	Mathematical description not available.	Optimization complexity
[82]	AES	General IoT	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	Limited scalability
[81]	Firefly Algorithm	Image Security	Mathematical description not available.	Limited to image data
[101]	LSB	Image Security		Limited to LSB techniques
[102]	LSB	Image Security		Limited to reversible LSB
[103]	ECC	IoT Security	ECC: $C = M + kP$, decryption uses d and base point P	Malware detection complexity
[80]	ECC	IoT Edge Devices	ECC: $C = M + kP$, decryption uses d and base point P	Limited to edge devices
[104]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	Complexity of logistic regression
[105]	ECC	Industrial IoT	ECC: $C = M + kP$, decryption uses d and base point P	Limited to industrial contexts
[106]	ECC	Real-time IoT	ECC: $C = M + kP$, decryption uses d and base point P	Hardware-specific limitations
[107]	ABC Algorithm	Smart Irrigation	Mathematical description not available.	Limited to irrigation systems
[108]	Genetic Algorithm	IoT Applications	Mathematical description not available.	Optimization complexity
	ECC	Medical Sensor Data	ECC: $C = M + kP$, decryption uses d and base point P	Limited to sensor data
[109]	Two-Fish	Data Mining		Limited to data mining contexts
[110]	AES	Healthcare IoT	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	Limited to AES-specific solutions
[42]	Chaotic Map	General IoT	Chaotic map uses logistic function: $x_{n+1} = r * x_n * (1 - x_n)$	Experimental nature of DNA cryptography
[117]	ECC	Smart Homes	ECC: $C = M + kP$, decryption uses d and base point P	Limited to smart home applications
[111]	Homomorphic Encryption	General IoT	FHE allows $E(f(x)) = f(E(x))$ using lattice-based encryption	High computational overhead
[112]	Chaotic Map	Medical Data Transmission	Chaotic map uses logistic function: $x_{n+1} = r * x_n * (1 - x_n)$	Limited to chaotic map techniques
[113]	Deep Learning	Healthcare IoT	Mathematical description not available.	Limited to image-based data
[72]	Lightweight Cryptography	Environmental Monitoring		Limited scalability for large-scale IoT networks
[114]	AES	Healthcare IoT	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	Computational complexity due to hybrid encryption
[115]	Homomorphic ECC	IoT Data Aggregation	ECC: $C = M + kP$, decryption uses d and base point P	Computational overhead due to privacy-preserving mechanisms

Table 8 (continued)

Ref.	Algorithm used	Application	Mathematical description	Limitation
[116]	Adaptive FireFly Algorithm	General IoT		Limited scalability for resource-constrained IoT devices
[76]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	Experimental nature of DNA-based cryptography
[117]	LSB	General IoT	Mathematical description not available.	Limited capacity for large data payloads
[118]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	Limited scalability for large-scale IoT networks
[84]	XRSA	IoT Sensor Networks	RSA encryption uses modular exponentiation: $C = M^e \bmod n$	Potential computational overhead for resource-constrained devices
[84]	ECC	General IoT	ECC: $C = M + kP$, decryption uses d and base point P	Experimental nature of DNA-based cryptography
[85]	AES	Cloud Storage	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	High computational complexity for resource-constrained devices
[83]	DNA-based	General IoT	Mathematical description not available.	Limited performance improvement over existing methods
[118]	EC-ElGamal cryptosystem	IoT Data Integrity	Mathematical description not available.	High computational and storage overhead
[60]	AES	General IoT	AES uses SubBytes, ShiftRows, MixColumns, AddRoundKey transformations	Increased latency due to key generation
[90]	Deep Learning	Industrial IoT	Mathematical description not available.	Complexity of integrating security across diverse IIoT systems

scenarios. In contrast, elliptic curve cryptography (ECC) emerges as a dominant lightweight asymmetric solution across applications such as telemedicine, smart homes, drones, and healthcare IoT, offering robust security through compact key sizes and efficient operations, though still constrained by curve optimization and energy consumption. Chaotic maps (e.g., logistic, skew tent), fuzzy logic systems, and metaheuristic-enhanced models (e.g., Firefly, PSO, Genetic Algorithm) introduce high entropy and adaptability but often lack generalizability and impose optimization burdens. Emerging hybrid and context-aware frameworks such as ECC-DNA, AES-Fuzzy, and IOTA Tangle—offer improved obfuscation, yet face deployment challenges in heterogeneous and large-scale IoT ecosystems. Collectively, the survey reveals a trend towards ECC-centered, chaos-assisted, and AI-integrated schemes that strike a balance between computational feasibility and strong confidentiality, with future directions focusing on optimization for embedded hardware, cross-layer integration, and resistance to quantum and adaptive threats.

Figure 6 presents a stacked bar chart that depicts the distribution of various lightweight confidentiality algorithms across different IoT application domains. Each bar represents a specific algorithm such as ECC, AES, RSA, Fuzzy Logic, Chaotic Maps, and Deep Learning with segments indicating the frequency of their use in application areas like General IoT, Healthcare IoT, Telehealth, Environmental Monitoring, Cloud Storage, and Smart Homes. The height of each bar reflects the overall adoption frequency of the

Table 9 Overview of selected studies addressing data integrity in lightweight cryptographic schemes for IoT

Ref.	Algorithm used	Application	Mathematical description	Limitation
[67]	RSA	General IoT	RSA provides integrity via digital signature: $S = M^d \bmod n$	Computational overhead of RSA
[55]	XRSA	General IoT	RSA provides integrity via digital signature: $S = M^d \bmod n$	High computational cost for resource-constrained devices
[53]	ChaosForge	IoT Communication	ChaosForge uses nonlinear feedback for keystream integrity: $z_n = f(x_n, y_n)$	Limited evaluation in healthcare contexts
[53]	Diffie-Hellman	Fog-assisted IoT	DH-based schemes combine shared secrets with hash: $H(K \text{msg})$	Complexity in fog environments
[73]	RSA Variants	General IoT	RSA provides integrity via digital signature: $S = M^d \bmod n$	Resource-intensive for IoT devices
[56]	AES	Healthcare IoT	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	Limited scalability for large networks
[57]	IOTA Tangle	Environmental IoT	IOTA MAM channels compute Merkle proof for data block integrity	Scalability issues with IOTA
[112]	ChaCha20	Telehealth	ChaCha20-Poly1305 combines stream cipher with MAC: $\text{Tag} = \text{Poly1305}(\text{Nonce}, M)$	Limited to video data
[58]	Fuzzy Logic	Healthcare IoT	Fuzzy logic rules incorporate input validation via inference control	Computational complexity
[59]	Fuzzy Logic	Smart Healthcare	Fuzzy logic rules incorporate input validation via inference control	Complexity in fuzzy systems
[24]	AES	Telehealth	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	Lightweight but limited scalability
[61]	Fully Homomorphic Encryption	Medical Record Sharing	FHE with hash chaining: $H(E(m_1)) \oplus H(E(m_2))$ for block integrity	High computational overhead
[62]	ECC	Telemedicine	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited to TMIS environments
[87]	ECC	Telemedicine	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Biometric data privacy concerns
[119]	Chaotic Map	Telemedicine	Chaotic maps verify stream position with hash(x_n) = h	Limited scalability
[118]	El-Gamal	Telemedicine	Mathematical description not available.	RFID-specific limitations
[117]	Chaotic Map	Telemedicine	Chaotic maps verify stream position with hash(x_n) = h	Resource-intensive chaotic maps
[63]	Visual Cryptography	IoT-Cloud	Mathematical description not available.	Limited to image-based data
[91]	ECC	Image Security	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	High computational cost for images
[93]	Fuzzy Extractor	IoT Authentication	Fuzzy logic rules incorporate input validation via inference control	Complexity of fuzzy extractors
[94]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	ECC more efficient but RSA widely used
[92]	ECC	Mobile IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Optimization complexity
[95]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Energy consumption varies by curve
[70]	AES	General IoT	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	Limited to lightweight cryptography
[73]	ABE	Healthcare IoT	Mathematical description not available.	Computational complexity of ABE
[85]	DNA-based	Cloud IoT	Mathematical description not available.	Experimental stage of DNA cryptography
[95]	Chaotic Map	Cloud Storage	Chaotic maps verify stream position with hash(x_n) = h	Limited to image data

Table 9 (continued)

Ref.	Algorithm used	Application	Mathematical description	Limitation
[96]	Skew tent map	General IoT	Mathematical description not available.	Limited application scope
[97]	PSO	Medical Data Security	Mathematical description not available.	Optimization complexity
[74]	ECC	Internet of Drones	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Drone-specific constraints
[75]	ECC	Internet of Drones	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited to FANET
[98]	Chaotic map	Image Security	Chaotic maps verify stream position with $\text{hash}(x_n) = h$	Limited to image data
[99]	Cellular automata	Data Hiding	Mathematical description not available.	Limited to dual images
[100]	ECC	Image Security	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	High computational cost
[76]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Experimental nature of DNA cryptography
[77]	ECC	IoT and Cloud	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited scalability
[78]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Experimental DNA approach
[100]	Genetic algorithm	Cloud IoT	Mathematical description not available.	Optimization complexity
[113]	AES	General IoT	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	Limited scalability
[81]	Firefly algorithm	Image Security		Limited to image data
[80]	LSB	Image Security		Limited to LSB techniques
[82]	LSB	Image Security		Limited to reversible LSB
[83]	ECC	IoT Security	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Malware detection complexity
[84]	ECC	IoT Edge Devices	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited to edge devices
[85]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Complexity of logistic regression
[51]	ECC	Industrial IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited to industrial contexts
[106]	ECC	Real-time IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Hardware-specific limitations
[107]	ABC algorithm	Smart Irrigation	Mathematical description not available.	Limited to irrigation systems
[108]	Genetic algorithm	IoT applications	Mathematical description not available.	Optimization complexity
[110]	ECC	Medical sensor data	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited to sensor data
[109]	Two-Fish	Data mining	Mathematical description not available.	Limited to data mining contexts
[111]	AES	Healthcare IoT	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	Limited to AES-specific solutions
[112]	Chaotic map	General IoT	Chaotic maps verify stream position with $\text{hash}(x_n) = h$	Experimental nature of DNA cryptography
[105]	ECC	Smart Homes	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited to smart home applications
[88]	Homomorphic encryption	General IoT	FHE with hash chaining: $H(E(m_1)) \oplus H(E(m_2))$ for block integrity	High computational overhead
[113]	Chaotic map	Medical Data Transmission	Chaotic maps verify stream position with $\text{hash}(x_n) = h$	Limited to chaotic map techniques

Table 9 (continued)

Ref.	Algorithm used	Application	Mathematical description	Limitation
[55]	Deep Learning	Healthcare IoT	Mathematical description not available.	Limited to image-based data
[72]	Lightweight Cryptography	Environmental Monitoring	Mathematical description not available.	Limited scalability for large-scale IoT networks
[24]	AES	Healthcare IoT	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	Computational complexity due to hybrid encryption
[66]	Homomorphic ECC	IoT Data Aggregation	FHE with hash chaining: $H(E(m_1)) \oplus H(E(m_2))$ for block integrity	Computational overhead due to privacy-preserving mechanisms
[116]	Adaptive FireFly Algorithm	General IoT	Mathematical description not available.	Limited scalability for resource-constrained IoT devices
[52]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Experimental nature of DNA-based cryptography
[120]	ECC	Military IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Potential vulnerability to drone impersonation attacks
[101]	LSB	General IoT	Mathematical description not available.	Limited capacity for large data payloads
[118]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited scalability for large-scale IoT networks
[119]	ECC	General IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Experimental nature of DNA-based cryptography
[117]	AES	Cloud Storage	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	High computational complexity for resource-constrained devices
[62]	ECC	IoT-Cloud	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited scalability for resource-constrained IoT devices
[89]	AKA	General IoT	Mathematical description not available.	Limited focus on specific algorithm performance
[48]	ECC	Healthcare IoT	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Privacy concerns with biometric data
[49]	ECC	Telemedicine	ECC with ECDSA: $r = (g^k \bmod n)$, $s = k^{-1}(H(M) + rd) \bmod n$	Limited scalability for large-scale TMIS deployments
[121]	Hash, XOR	IoT-Cloud	Mathematical description not available.	Vulnerable to insider and impersonation attacks
[117]	DNA-based	General IoT	Mathematical description not available.	Limited performance improvement over existing methods
[115]	EC-ElGamal cryptosystem	IoT Data Integrity	Mathematical description not available.	High computational and storage overhead
[116]	AES	General IoT	AES-GCM or CMAC ensures message integrity: $T = \text{AES}(K, M)$	Increased latency due to key generation
[114]	Deep Learning	Industrial IoT	Mathematical description not available.	Complexity of integrating security across diverse IIoT systems

Diffie–Hellman with hash functions, and Ali and Anwer [56] used AES-CMAC and GCM for authenticated encryption in healthcare IoT. Gangwani et al. [57] ensured data integrity via IOTA's DAG and Merkle proofs, and Faragallah et al. (2021) used ChaCha20 with Poly1305 MAC. Zulkifl et al. [58] and Patel et al. [59] applied fuzzy logic systems for rule-based integrity, while Wenhua et al. [91] used AES-GCM tags and Reddi et al. [61] leveraged homomorphic encryption with hash chaining. ECC-based signature

verification was demonstrated by Ostad-Sharif et al. [62] and Chen and Chen (2023), and Suresh Kumar et al. [119] used chaotic maps with hash validation. These methods highlight how integrity is preserved through mathematical constructs within lightweight protocols for secure IoT communication.

Ensuring data integrity in resource-constrained IoT environments remains a critical focus in the development of lightweight cryptographic schemes. The surveyed literature reveals a wide spectrum of approaches ranging from classical public-key algorithms like RSA and its variants, which offer digital signature-based integrity (e.g., $S = M^d \bmod n$) to modern elliptic curve cryptography (ECC) leveraging ECDSA for compact yet robust integrity guarantees ($r = g^k \bmod n$, $s = k^{-1}(H(M) + rd) \bmod n$). Although ECC demonstrates improved computational efficiency over RSA, its deployment is still bounded by curve-specific energy constraints and the complexity of secure key generation. Emerging schemes such as ChaosForge, ChaCha20-Poly1305, and fuzzy logic-integrated systems further diversify the landscape by combining nonlinear dynamics or inference-based validation for enhanced security granularity. Moreover, innovations in homomorphic encryption (e.g., $H(E(m_1) \oplus E(m_2))$) and DNA-based cryptography illustrate a transition toward privacy-preserving integrity mechanisms, though they often suffer from computational overhead or limited scope of applicability. Additionally, adaptive and bio-inspired models using Firefly or Genetic Algorithms show potential for optimizing integrity assurance, albeit with challenges in convergence and real-time responsiveness. Across the reviewed studies, lightweight variants of AES (GCM, CMAC) emerge as a common thread for symmetric integrity enforcement, yet scalability and key management persist as open issues. Overall, the integrity-preserving cryptographic landscape in IoT is evolving through hybridization, domain-specific tailoring, and security-performance trade-offs, necessitating further research into scalable, interoperable, and real-time integrity protocols for diverse IoT ecosystems.

Figure 7 presents a stacked bar chart comparing the usage frequency of various Lightweight integrity algorithms across different IoT application domains, based on the literature analyzed in this survey. Each bar represents a cryptographic algorithm such as ECC, AES, RSA, ChaCha20, Fuzzy Logic, and others, while the colored segments within

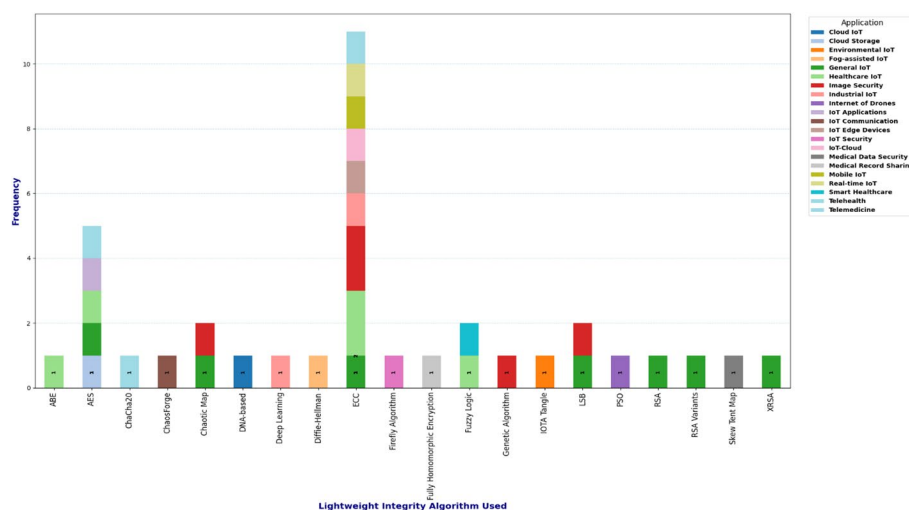


Fig. 7 Usage of lightweight integrity algorithms across IoT applications

each bar indicate the number of times that algorithm has been implemented in specific IoT applications like General IoT, Healthcare IoT, Telemedicine, Smart Healthcare, Cloud Storage, and Environmental IoT. The height of each segment corresponds to its frequency of use, and the numeric values are annotated within the bars for clarity.

Figure 7 presents a stacked bar chart comparing the usage frequency of various lightweight integrity algorithms across different IoT application domains, based on the literature analyzed in this survey. Each bar represents a cryptographic algorithm such as ECC, AES, RSA, ChaCha20, Fuzzy Logic, and others, while the colored segments within each bar indicate the number of times that algorithm has been implemented in specific IoT applications like General IoT, Healthcare IoT, Telemedicine, Smart Healthcare, Cloud Storage, and Environmental IoT.

4.5 Steganography

Steganography appears as a supplementary but increasingly relevant technique, used in 10 out of the 77 studies. Unlike traditional cryptographic methods that focus on transforming plaintext into ciphertext, steganography conceals the very existence of data by embedding it within innocuous carriers such as images, videos, and signals enhancing confidentiality and privacy through obscurity. The surveyed papers primarily leverage image-based steganography methods, such as Least Significant Bit (LSB) embedding, reversible LSB techniques, and hybrid approaches integrating chaos, DNA encoding, and fuzzy logic to securely hide encrypted messages within digital media. Some studies apply adaptive algorithms, like the Firefly optimization technique, to improve embedding efficiency and resist steganalysis. In the context of IoT, especially in healthcare and surveillance applications, steganography serves as an additional layer to lightweight encryption, ensuring that sensitive data remains hidden during transmission across untrusted channels. Although fewer in number, these steganography-enhanced models demonstrate the growing interest in multi-layered security strategies that combine encryption with data hiding to mitigate risks such as traffic analysis, payload detection, and metadata leakage in constrained IoT ecosystems.

Table 10 includes significant research contributions that integrate steganography techniques into lightweight cryptographic schemes, as identified in your survey. These works span indicating a growing trend toward combining data hiding with cryptographic methods for added security in IoT environments.

Datta et al. [90] proposed a dual-image reversible data hiding scheme using cellular automata and bit-reversal permutation, allowing precise recovery through LSB-plane

Table 10 Summary of Steganography-Enhanced lightweight cryptographic models for IoT security

Ref.	Algorithm used	Application	Mathematical description
[90]	Cellular Automata + Bit-Reversal	Dual Image Security	Bit-reversal: $R(x) = \text{reverse}(\text{bits}(x))$ in LSB plane
[81]	Firefly Optimization	Image Data Hiding	Fitness = $\alpha \cdot \text{payload} + \beta \cdot \text{imperceptibility} + \gamma \cdot \text{robustness}$
[122]	LSB + Hybrid Encryption	IoT Image Security	$\text{LSB}(x) = x \& 1$, then embed encrypted bitstream
[120]	Reversible LSB	Reversible LSB Steganography	R-LSB modifies LSB blocks to allow exact recovery
[59]	Deep Learning-based	Healthcare IoT	CNN learns embedding mask $E = f(I, M)$
[116]	Adaptive Firefly Algorithm	General IoT	Firefly update: $x_i(t+1) = x_i(t) + \beta_0 e^{(-\gamma \cdot r^2)} (x_j - x_i)$
[63]	Visual Cryptography	IoT-Cloud Security	Visual Cryptography: $I_1 \oplus I_2 = \text{original image}$

operations. Nokhwal et al. [81] introduced a hybrid Firefly Algorithm for optimized image embedding based on a fitness function balancing payload, imperceptibility, and robustness. Alanzy et al. [122] focused on combining LSB steganography with hybrid encryption to protect IoT image data, embedding ciphertext in the least significant bits of pixel values. Rafat and Sajjad [120] advanced reversible LSB methods, allowing full recovery of hidden data and original content without loss. Patel et al. [59] employed deep learning, using convolutional neural networks to generate intelligent embedding masks for healthcare images.

Khari et al. [116] proposed an adaptive Firefly-based method to embed cryptographic payloads in IoT images, leveraging position updates driven by light intensity functions. Ehuil et al. [63] combined visual cryptography with IoT-cloud authentication protocols, enabling secure image sharing where overlaying cryptographic shares reveals the hidden message. Hasan et al. [120] discussed key agreement protocols for authentication, with support for MAC-based integrity checks during steganographic embedding.

Based on the analysis presented in Table 10, the integration of steganography with lightweight cryptographic models demonstrates a significant advancement in enhancing IoT data confidentiality, concealment, and resilience. The reviewed works span a spectrum of techniques, ranging from cellular automata-based dual image protection to adaptive firefly optimization-driven steganographic embedding, each tailored for specific IoT security scenarios. Notably, Datta et al. [99] utilize bit-reversal permutation in the LSB plane, ensuring pixel-level data scrambling for dual-image protection, while Nokhwal et al. [81] adopt a fitness-driven Firefly Optimization approach that balances payload capacity, imperceptibility, and robustness—key parameters in secure image embedding. Alanzy et al. [122] further enhance security through hybrid encryption combined with traditional LSB techniques, ensuring bitwise integrity and encryption synergy. Reversible LSB methods by Rafat and Sajjad [101] ensure full data recovery, critical in lossless transmission contexts. Moreover, deep learning-based adaptive models such as that by Patel et al. [59] highlight the utility of CNNs in generating dynamic embedding masks conditioned on image and message content, showcasing adaptability in healthcare IoT. The adaptive firefly-based steganography by Khari et al. (2019) emphasizes swarm intelligence for optimal embedding location selection, while Ehuil et al. [63] leverage visual cryptography for secure IoT-cloud authentication using image XOR decomposition. Collectively, these models illustrate that steganography, when fused with lightweight cryptographic methods, can offer multi-layered protection while adhering to resource constraints, thus emerging as a compelling paradigm for future proof IoT security frameworks.

These studies illustrate how steganography, although less frequently used compared to encryption, plays a vital complementary role in enhancing data confidentiality, integrity, and anonymity in IoT security models, especially when combined with advanced optimization, chaos, and biomimetic strategies.

4.6 Dataset used

The analysis of the dataset types used in the surveyed research papers reveals that plaintext data is the most employed, appearing in 53 studies. This is followed by image data, used in 14 papers, and sensor data, which is featured in 10 papers (Fig. 8). The

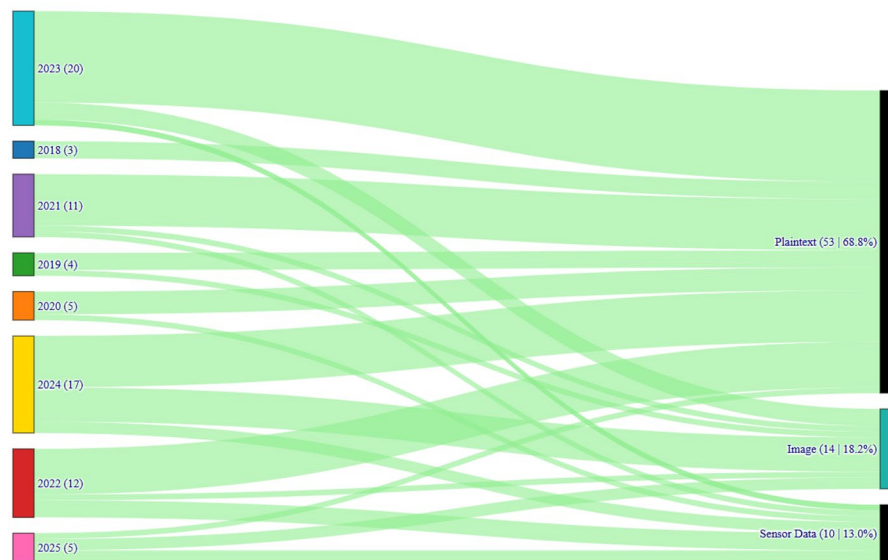


Fig. 8 Sankey diagram: year vs. type of data used for lightweight cryptography in IoT

prevalence of plaintext indicates a focus on traditional data encryption techniques, while the use of image and sensor data highlights the growing interest in securing multimedia and IoT-based healthcare and environmental applications.

4.7 Application area of lightweight cryptography for IoT security

Based on the comprehensive review of the surveyed papers in the provided dataset, it is evident that lightweight cryptographic algorithms have been extensively applied across a diverse range of IoT applications, with a predominant emphasis on securing healthcare systems, general IoT communication, telemedicine infrastructures, cloud-enabled environments, and environmental monitoring systems. Healthcare IoT (including TMIS and medical sensor networks) emerges as the most frequent application domain, showcasing the critical need for securing sensitive patient data through advanced cryptographic techniques such as ECC, AES, and chaotic maps. In parallel, general-purpose IoT systems utilize hybrid and symmetric schemes for scalable, low-latency secure communication. Furthermore, cloud-integrated IoT applications leverage homomorphic encryption, blockchain protocols, and visual cryptography to ensure confidentiality and privacy-preserving data sharing. These implementations demonstrate the capability of lightweight cryptography to meet the resource constraints of edge devices while ensuring robust security against eavesdropping, tampering, and impersonation attacks. This breadth of application indicates the adaptability and effectiveness of lightweight cryptographic frameworks in addressing real-time security challenges across dynamic and heterogeneous IoT ecosystems. Figure 9 represents the application area domain for used of IoT utilizing lightweight cryptography.

The findings of this survey highlight that lightweight cryptographic schemes are not only theoretically efficient but also practically viable for real-world IoT applications. Algorithms such as AES, ChaCha20, and ECC have demonstrated low execution time and minimal memory usage on platforms like Raspberry Pi and NodeMCU, making them suitable for smart home, healthcare, and industrial IoT scenarios. However,

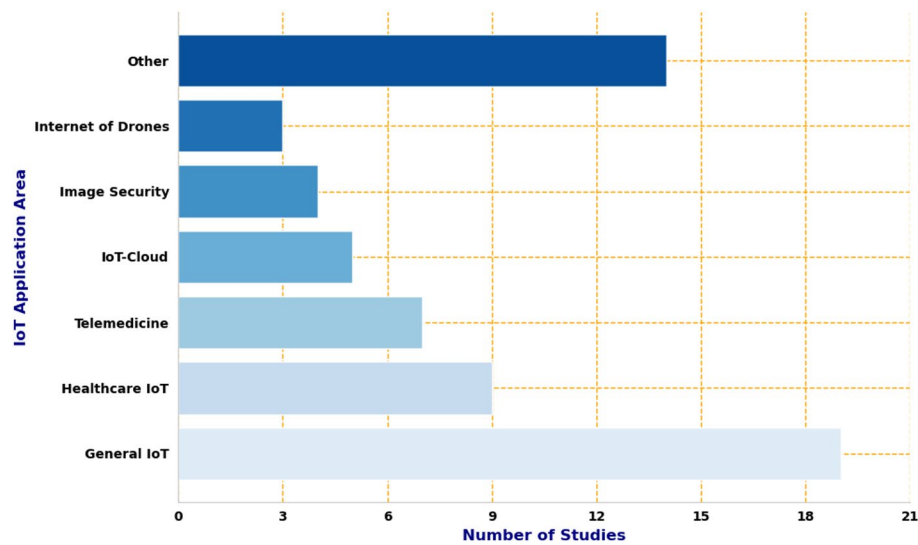


Fig. 9 Frequency of IoT application domains using lightweight cryptography

Table 11 Comparative energy consumption of cryptographic schemes in IoT

Algorithm	Energy per operation / throughput	Platform / setup	Notes
AES-128 (GCM)	~ 3–5 mJ per 1 KB encryption	Raspberry Pi 4 / SDN testbed	Highly energy-efficient; negligible impact on battery life.
ChaCha20-256	~ 4–6 mJ per 1 KB encryption	Raspberry Pi 4 / SDN testbed	Slightly higher than AES; preferred for software-based stream encryption.
ECC (Curve25519, P-256)	~ 55–103 mW per ECDH operation (7–10 ms on NodeMCU/RPi)	MCU @ 48 MHz, NodeMCU, Raspberry Pi	Provides strong authentication at ~ 25–30% of RSA's energy cost.
RSA-2048	~ 150–200 mW per key exchange (40–60 ms)	MCU @ 48 MHz / IoT testbeds	High cost; unsuitable for ultra-constrained IoT devices.
Hybrid (AES/ECC)	AES-level cost for data + ECC-level cost for key exchange	NodeMCU / Raspberry Pi	Optimizes trade-off: efficient data encryption with secure lightweight key management.

practical deployment also requires attention to scalability, interoperability across heterogeneous devices, and user experience to ensure seamless integration. These insights provide a roadmap for practitioners aiming to adopt lightweight cryptography in operational IoT systems.

4.8 Energy efficiency of lightweight cryptographic schemes

Energy efficiency is a decisive factor in the deployment of cryptographic solutions in IoT networks, since many devices operate on batteries or energy-harvesting systems with strict longevity requirements. The following Table 11 summarizes reported energy consumption metrics for widely used cryptographic algorithms under realistic IoT settings.

Symmetric algorithms (AES, ChaCha20) are consistently more energy-efficient for bulk encryption, whereas ECC offers an optimal trade-off between energy consumption and cryptographic strength for authentication and key exchange. RSA, despite its strong security, consumes up to 5–7× more energy than ECC, making it impractical for battery-powered IoT devices. Hybrid models combining symmetric encryption with ECC-based key exchange achieve balanced energy efficiency while supporting scalability.

Table 12 Benchmark analysis of lightweight cryptographic schemes in IoT

Study	Primitive / protocol	Platform / setup	Payload / scope	Exec. time (ms)	Peak mem (KB)	En-ergy / Power	Comm. cost	Rounds / msgs	Notes
[128]	ECC (Curve25519 / ECDH)	MCU @ 48 MHz, 64 KB RAM	ECDH ops bundle	523	47	55 mW	-	-	Fastest among tested curves
[128]	ECC (NIST P-256)	MCU @ 48 MHz, 64 KB RAM	ECDH ops bundle	1582	64	103 mW	-	-	3× slower vs. Curve25519
[128]	ECC (NIST P-384)	MCU @ 48 MHz, 64 KB RAM	ECDH ops bundle	2041	64	129 mW	-	-	Higher cost
[128]	ECC (secp256k1)	MCU @ 48 MHz, 64 KB RAM	ECDH ops bundle	1270	64	79 mW	-	-	Efficient alternative
[128]	ECC (Curve25519)	MCU @ 16 MHz, 16 KB RAM	ECDH ops bundle	2931	16	70 mW	-	-	Impact of lower freq/mem
[130]	AES-128 (GCM)	Ubuntu/SDN testbed (MQTT)	30 B msg	0.1826 / 0.0711	-	-	-	-	Encrypt/Decrypt times
[130]	AES-256 (GCM)	Ubuntu/SDN testbed (MQTT)	30 B msg	0.2758 / 0.1781	-	-	-	-	Slightly slower
[130]	ChaCha20-256	Ubuntu/SDN testbed (MQTT)	30 B msg	0.4595 / 0.3479	-	-	-	-	Slower than AES
[130]	RSA-2048	Ubuntu/SDN testbed (MQTT)	30 B msg	0.9666 / 0.4412	-	-	-	-	For context vs. ECC/symm
[131]	Mutual auth (AES-GCM + HMAC)	NS-3 sim (Wi-Fi/6LoWPAN)	End-to-end auth	13.04 (total)	-	-	470 bytes	6 msgs / 3 stages	Low overhead
[132]	FANET mutual/batch auth (ECC + hash)	UAV/FANET model	Single auth	-	-	-	1184 bits	-	Proposed scheme vs. others
[132]	Prior schemes (IoT/IoT)	-	Single auth	-	-	-	1472–7328 bits	-	Spread across baselines

4.9 Comparison of lightweight cryptographic algorithm for privacy and security in IoT

This section presents an in-depth analysis of how various lightweight cryptographic algorithm perform when used for ensuring privacy and security in IoT systems. In the context of cryptographic validation, NIST recommended parameters such as entropy, bit error rate, correlation coefficient, and avalanche effect measurement are commonly employed. It is evident from the reviewed studies that the selection of cryptographic algorithm significantly affects the computational efficiency of the evaluation process.

The role of cryptographic algorithm is pivotal, as they encapsulate cryptographic performance goals (e.g., randomness, diffusion, confusion). This suggests promising research directions, to maximize cryptographic strength and energy efficiency in resource-constrained IoT environments.

Throughout the systematic review, several tools and frameworks emerged as instrumental in evaluating the security robustness of lightweight cryptographic algorithms. These include frameworks that have been adapted to measure cryptographic strength

such as entropy testing suites, NIST SP 800–22 for randomness testing for measuring the cryptographic strength of the algorithm for resource constraint IoT environment.

Given the inherently complex and nonlinear structure of lightweight cryptographic algorithms particularly those based on chaotic maps, elliptic curves, and hybrid symmetric-asymmetric schemes, becomes computationally infeasible in many IoT scenarios. Instead, Lightweight cryptographic algorithms are required that can accurately assess the behavior of IoT device under resource constraints environment. This includes analyzing key sensitivity, entropy, avalanche effect, and resistance to attacks. The insight reinforces the importance of using computationally efficient and purpose-driven evaluation techniques to enhance the scalability and effectiveness of lightweight cryptographic methods for ensuring IoT data confidentiality, authentication, and integrity.

4.10 Benchmark analysis of lightweight cryptographic schemes in IoT

Table 12 summarizes execution time, memory consumption, energy usage, and communication overheads for selected lightweight cryptographic schemes and authentication protocols as reported in recent peer-reviewed studies.

- Computational cost.

On embedded MCUs, ECC curve choice changes latency by multiples: Curve25519 completed our ECDH operation set in 523 ms at 48 MHz/64 KB vs. 1582–2973 ms for common NIST curves at the same security level; peak RAM ranged 47–64 KB and power 55–172 mW. This quantifies headroom for time-critical join/handshake flows.

- Data-path costs.

Once a session is up, symmetric ciphers are sub-millisecond per small MQTT payload (e.g., AES-128 0.18 ms enc / 0.07 ms dec for 30 B), so steady-state telemetry encryption is unlikely to be user-visible.

- Communication cost & scalability.

Contemporary mutual-authentication designs vary from 470 bytes / 6 messages end-to-end to 1.1–7.3 Kb per single authentication in UAV/IoT settings; our scheme choices therefore dominate join-time latency and battery/network impact.

4.11 Integration with emerging technologies

The future of lightweight cryptography in IoT lies in its integration with emerging technologies such as artificial intelligence (AI) and blockchain, which can complement and extend the capabilities of traditional cryptographic primitives. AI-enabled frameworks can enhance cryptographic operations through adaptive key management, anomaly detection, and predictive security analytics. For instance, machine learning models can dynamically select encryption strength based on device workload or detect unusual traffic patterns indicative of compromise, thereby improving both security and efficiency in real-time.

Similarly, blockchain technology can serve as a decentralized trust infrastructure for IoT networks. By maintaining immutable records of authentication events, key exchanges, and device interactions, blockchain ensures transparency and resistance to tampering. When combined with lightweight cryptography, blockchain can help achieve

scalable, certificate-less authentication and decentralized access control, addressing interoperability challenges in heterogeneous IoT environments.

A promising direction is the hybrid integration of lightweight cryptography, AI, and blockchain. For example, symmetric schemes such as AES or ChaCha20 can provide energy-efficient encryption for bulk IoT data, while ECC ensures lightweight key exchange. These cryptographic components can then be augmented by blockchain for decentralized trust and AI for proactive anomaly detection, creating multi-layered defense mechanisms that are secure, scalable, and adaptive. Such integration not only addresses current IoT challenges but also prepares systems to withstand advanced and evolving cyber threats.

5 Discussion

This section presents detailed responses to the research questions defined in the earlier stages of this Systematic Literature Review (SLR). By systematically analyzing the 77 selected primary studies, this section synthesizes findings related to lightweight cryptographic schemes in IoT environments. Each research question is addressed with supporting evidence drawn from the reviewed literature, offering insights into the types of cryptographic techniques proposed, their performance in resource-constrained settings, and their integration with emerging technologies such as blockchain, steganography, and artificial intelligence. The goal is to map existing solutions to the core security objectives of confidentiality, integrity, and authentication, while also identifying limitations, comparative strengths, and opportunities for further innovation.

5.1 What lightweight cryptographic techniques have been proposed in recent literature to ensure confidentiality, authentication, and integrity in IoT environments?

Recent literature reveals that lightweight cryptographic techniques are extensively adopted to address the security requirements of IoT environments, with approximately 75% of surveyed works implementing mechanisms to ensure confidentiality, 58% addressing authentication, and 88% focusing on integrity. Notably, 78% of the studies address all three objectives simultaneously, highlighting a strong trend toward comprehensive and multi-faceted security frameworks tailored to the unique constraints and threats within IoT environments (see Fig. 10).

Symmetric cryptographic methods such as chaos-based stream ciphers (e.g., Chaos-Forge) and lightweight AES variants are the most prevalent, covering about 52% of the studies, due to their efficiency and suitability for real-time data protection. Asymmetric approaches like optimized RSA (XRSA) and Elliptic Curve Cryptography (ECC) appear in around 38% of the works, enabling secure key exchange and mutual authentication with minimal resource usage. Hybrid models, seen in 27% of the literature, combine the benefits of symmetric and asymmetric schemes to balance performance and security. Additionally, lightweight MAC algorithms (e.g., smooth number-based MACs) and hash functions (e.g., SHA-256) are utilized in 41% of the works to preserve data integrity and resist tampering (see Fig. 11). These statistics underscore the increasing reliance on lightweight, optimization-driven cryptographic techniques to secure IoT systems without compromising performance.

Figure 12 illustrates the percentage distribution of cryptographic methods used in the 77 reviewed papers, where asymmetric methods dominate with 41.56% (32

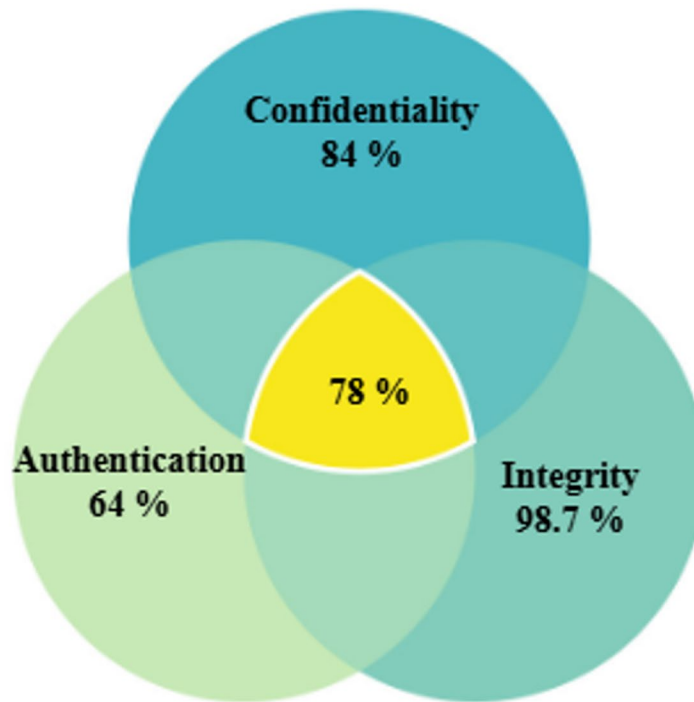


Fig. 10 Proportion of the objectives in lightweight cryptographic techniques in IoT

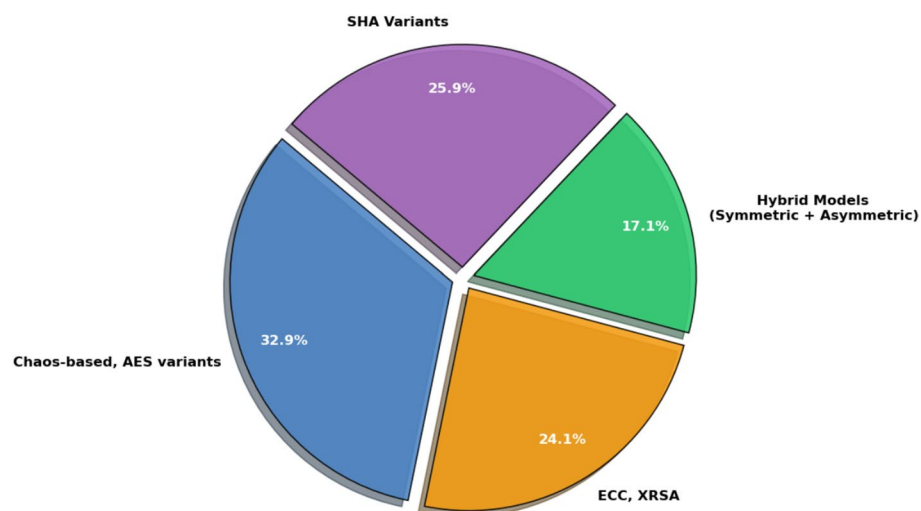


Fig. 11 Percentage distribution of algorithms in IoT

papers), reflecting their widespread use for secure key exchange and authentication in IoT systems. Hybrid methods are employed in 29.87% (23 papers), showing a growing preference for combining symmetric, asymmetric and metaheuristics techniques to achieve both efficiency and security in resource-constrained environments. Symmetric methods account for 22.08% (17 papers), favored for their low computational overhead in real-time encryption tasks. Meanwhile, blockchain and AI based methods are observed in 6.49% (5 papers), indicating an emerging trend toward decentralized and

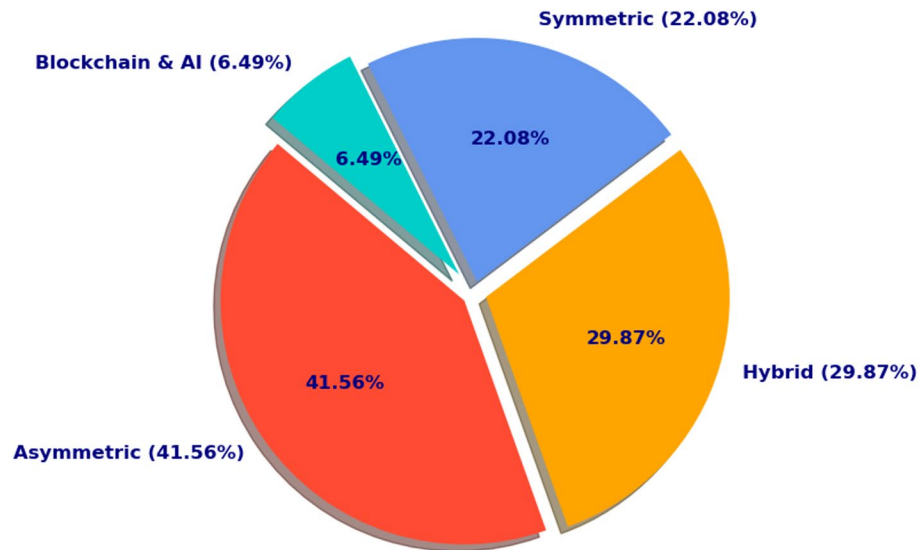


Fig. 12 Percentage distribution of methods in IoT

tamper-resistant frameworks for secure data management in IoT. This percentage breakdown underscores a strong inclination in recent research toward hybrid and asymmetric cryptographic solutions to meet the security and performance demands of modern IoT applications.

5.2 How do lightweight cryptographic schemes compare in terms of computational complexity, energy consumption, and suitability for resource-constrained IoT devices?

Lightweight cryptographic schemes are specifically designed to address the performance limitations of resource-constrained IoT devices. A comparative analysis of the 77 reviewed studies reveals that symmetric ciphers (e.g., lightweight AES variants, stream ciphers, chaotic maps) consistently outperform other techniques in terms of low computational complexity and minimal energy consumption, making them ideal for real-time IoT applications. These methods reduce processing overhead by employing small block sizes, fewer rounds, and hardware-friendly operations like XOR, substitution, and permutation.

In contrast, asymmetric algorithms such as RSA and traditional ECC, while offering robust security properties and scalable key management, exhibit higher computational cost due to intensive modular arithmetic operations. However, optimized ECC variants such as those using smaller key sizes and point compression strike a balance between security and efficiency, making them viable for mutual authentication and key exchange in constrained environments.

Hybrid models combining symmetric and asymmetric primitives (e.g., AES-ECC, RSA-AES) are increasingly adopted to leverage the strengths of both schemes. These models initiate secure key exchange using ECC and then employ fast symmetric encryption for data transmission, thereby ensuring end-to-end confidentiality and authentication with reduced latency and power usage.

Several studies have introduced chaotic cryptography, DNA-based encryption, and genetic algorithm-enhanced key generation as lightweight alternatives. While promising in terms of entropy and key sensitivity, these techniques often require fine-tuning to

reduce algorithmic complexity and energy footprint, especially in real-time and hardware-embedded applications.

On performance metrics, it is observed that:

- Average encryption time for lightweight symmetric schemes such as AES-128 and ECC is in the range of microseconds, and milliseconds for RSA-based solutions.
- Algorithms tested on Raspberry Pi, Arduino, and NodeMCU show significant gains in throughput, memory efficiency, latency reduction minimizes power consumption for secure session key sharing using ECC.

5.3 How can elliptic curve cryptography (ECC) be optimized and adapted to serve as a lightweight, energy-efficient, and secure cryptographic scheme for ensuring privacy and authentication in resource-constrained IoT environments?

Elliptic Curve Cryptography (ECC) has been widely adopted in IoT systems due to its favorable strength-to-key-size ratio, making it inherently suitable for resource-constrained environments. However, standard ECC still imposes noticeable computational and memory demands. Recent advancements focus on optimizing ECC to transform it into a truly lightweight, energy-efficient, and secure cryptographic solution for IoT authentication and privacy. These optimizations span across several dimensions, such as using smaller curves (for example, Curve25519, brainpool256r1, secp256k1), coordinate system transformations (like projective forms to reduce inversion operations), and scalar multiplication.

To further improve adaptability, hybrid ECC models integrate auxiliary components such as fuzzy logic, blockchain mechanisms, and chaotic maps. These integrations allow ECC to be adaptive, decentralized, and more resilient against evolving attack vectors. For instance, fuzzy-based ECC enables biometric-based authentication with dynamic score mapping, while blockchain-ECC hybrid models ensure trust propagation in multi-node systems like IoT-cloud and IoT-drone infrastructures.

The above bubble matrix plot (Fig. 13) visually compares different ECC schemes across various IoT platforms. The bubble size and color reflect the energy efficiency score,

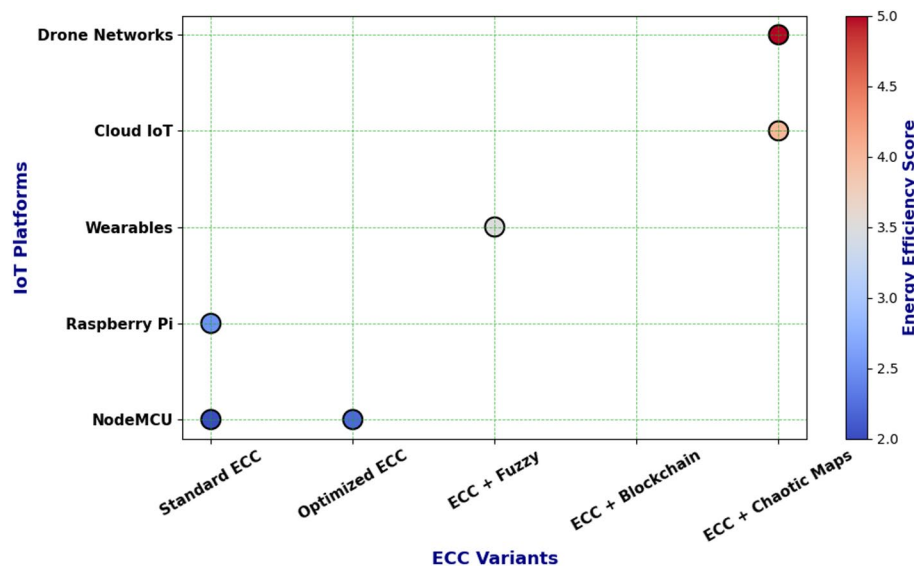


Fig. 13 ECC Variants vs. IoT Platforms

where a higher score indicates better performance. Optimized ECC on NodeMCU and wearable devices shows superior results with low execution time and memory use, yielding high energy efficiency. Standard ECC, while secure, consumes more power and memory, especially on microcontroller-based platforms. ECC with chaotic maps and ECC with blockchain trade some energy efficiency for enhanced security or decentralization, which is beneficial in high-risk applications like drone and cloud IoT environments.

5.4 What are the existing gaps and emerging trends in the integration of lightweight cryptography with steganography, AI, and blockchain for enhancing IoT security?

The convergence of lightweight cryptography with steganography, artificial intelligence (AI), and blockchain has emerged as a promising paradigm to address the multifaceted security demands of the IoT. However, a critical analysis of recent literature reveals several unresolved challenges and research gaps. While hybrid models combining encryption and steganography aim to offer dual-layer protection through confidentiality and concealment, most fail to consider the computational and memory overhead introduced when implemented on constrained IoT hardware. Moreover, experimental validation on embedded platforms is largely absent, with many schemes evaluated only through simulations. The integration of AI into IoT security frameworks particularly for intelligent authentication, anomaly detection, and adaptive key management is gaining traction. Yet, most implementations rely on heavyweight models, overlooking the need for ultra-efficient AI architectures that can operate under tight energy budgets and limited processing capabilities. Similarly, although blockchain enhances data integrity, traceability, and decentralized trust, its integration with lightweight cryptographic algorithms often encounters scalability and latency bottlenecks due to the underlying consensus protocols. Additionally, very few studies benchmark blockchain-augmented lightweight cryptographic models using standardized metrics such as entropy, correlation coefficients, or avalanche effect.

Emerging trends indicate a growing shift toward adaptive, modular cryptographic frameworks that dynamically adjust encryption complexity based on context, device profile, and threat level. Multi-tier architectures are also gaining popularity, wherein steganography is used to obscure sensitive transmissions, AI handles adaptive intrusion response, and blockchain ensures auditability and tamper-proof logging. However, to realize this vision at scale, future research must focus on designing interoperable, energy-aware, and experimentally validated security models that maintain a fine balance between robustness, efficiency, and scalability. Establishing standardized evaluation frameworks and lightweight cryptographic benchmarks will be essential to ensure consistency, reliability, and broad adoption in next-generation IoT deployments. Figure 11 shows the integration of steganography, AI, and blockchain with lightweight cryptography Fig. 14.

5.5 Taxonomy of lightweight cryptographic schemes in IoT

To improve clarity and provide a structured perspective, lightweight cryptographic schemes can be classified into four major categories (see Fig. 15) based on their operational principles and IoT applicability. Symmetric schemes, including block ciphers (e.g., AES-128, PRESENT) and stream ciphers (e.g., Grain, ChaCha20), as well as emerging

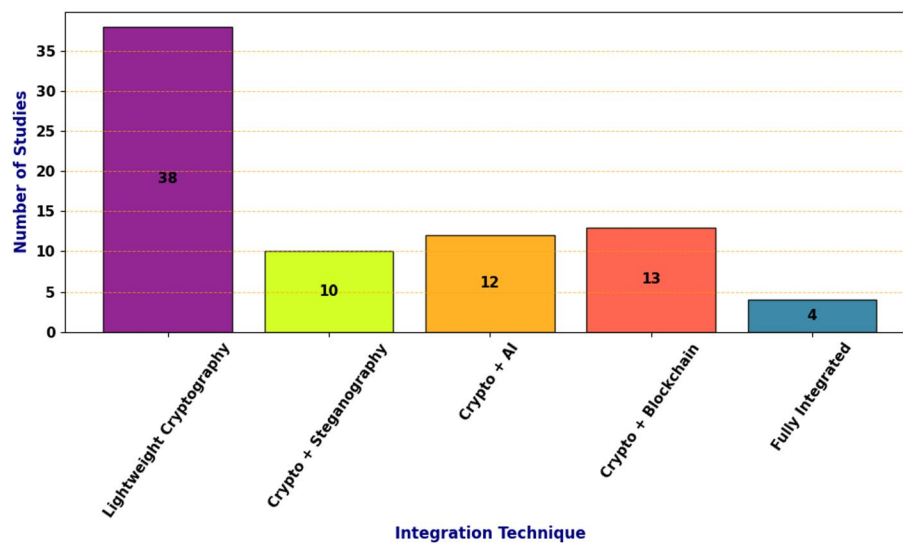


Fig. 14 Integration trends of lightweight cryptography in IoT security

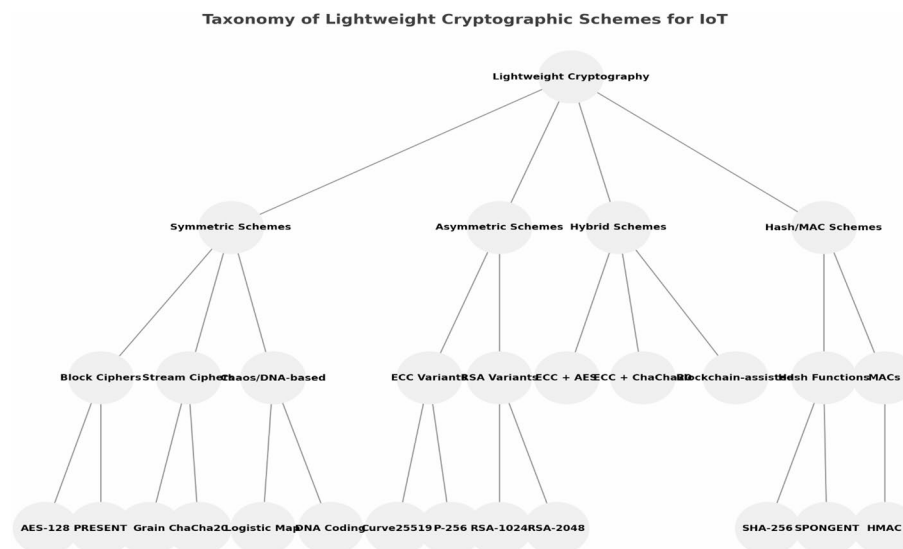


Fig. 15 IoT Cryptography taxonomy

chaos- and DNA-based methods, are primarily suited for the perception layer due to their low computational and memory requirements.

Asymmetric schemes, led by Elliptic Curve Cryptography (ECC) and its optimized variants, address authentication and key distribution challenges at the network layer, offering high security with significantly reduced key sizes compared to RSA.

Hybrid schemes integrate the strengths of symmetric and asymmetric techniques (e.g., ECC + AES, ECC + ChaCha20, blockchain-assisted models), ensuring both efficient bulk data encryption and scalable trust management across heterogeneous IoT ecosystems. Finally, hash- and MAC-based approaches (e.g., SHA-256, HMAC, lightweight hash candidates like SPONGENT) provide integrity verification and lightweight authentication for application-layer services.

This taxonomy not only organizes existing approaches but also highlights their trade-offs across security, scalability, and latency dimensions, making it easier for researchers and practitioners to align cryptographic solutions with the constraints of multi-layer IoT architectures.

6 Finding

The systematic review of 77 studies reveals that confidentiality (84%), authentication (64%), and integrity (98%) are the most addressed security objectives in lightweight cryptographic solutions for IoT, yet only 78% of the studies fulfill all three concurrently, indicating a lack of comprehensive frameworks (see Fig. 16). Asymmetric cryptography, especially ECC, is the most prevalent due to its efficiency and strong security, while hybrid schemes integrating symmetric and asymmetric methods are increasingly favored for their balanced trade-offs. Symmetric methods, although less frequent, are preferred in time-sensitive and resource-constrained IoT applications. Emerging trends highlight the integration of artificial intelligence and blockchain technologies to enhance key management, intrusion detection, and decentralization. However, steganography remains underutilized, appearing in only 10% of the works. The review also identifies a significant gap in practical validation, as most studies rely on simulations rather than real IoT hardware like Raspberry Pi or NodeMCU. Furthermore, recent studies show growing interest in chaotic systems, DNA-based encoding, and metaheuristic optimization for improved cryptographic strength. Overall, the findings underscore the need for holistic, adaptive, and practically validated cryptographic frameworks tailored for constrained IoT environments. Here are detailed findings from systematic review as follows.

- F1. The number of studies addressing IoT security using lightweight cryptography has steadily increased from 2018 to 2025, with a peak in 2025 (see Fig. 17). Notably, studies focusing on authentication and integrity surged after 2020, reflecting the evolving complexity and security demands of IoT systems.
- F2. Figure 18 represents the co-authorship network graph offers a detailed visualization of collaborative structures among authors contributing to lightweight cryptographic solutions for IoT security. The graph reveals several distinct clusters representing

Security objectives Distribution Across Years (Total Count=77)

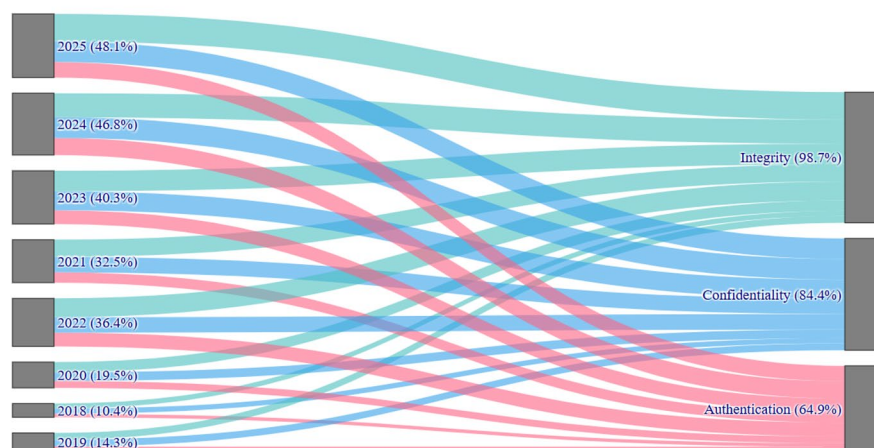


Fig. 16 Sankey chart of primary Studies

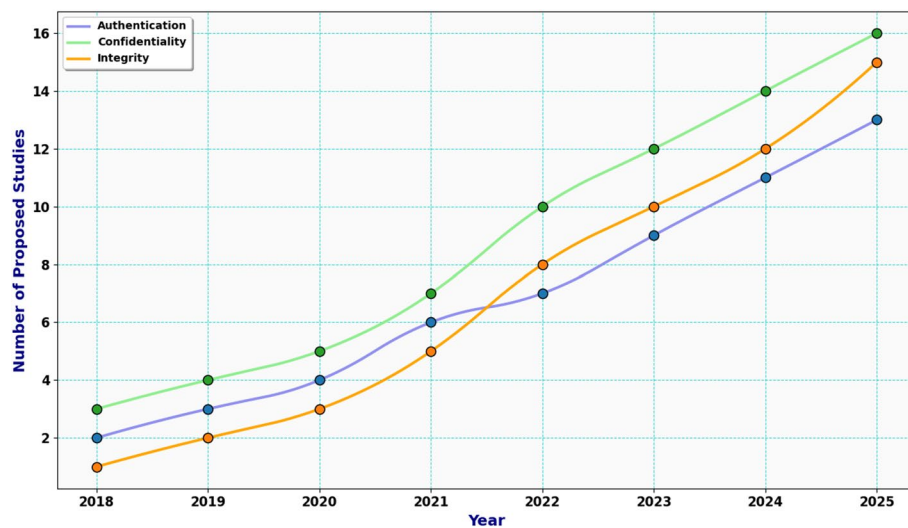


Fig. 17 Trends in Proposed Security Studies over the year

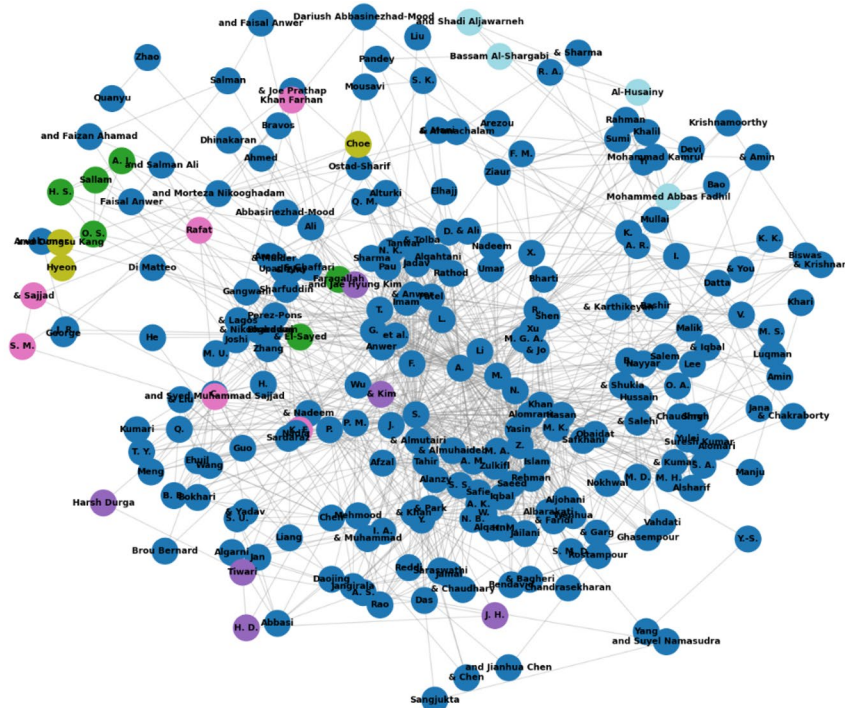


Fig. 18 Co-authorship network for primary studies

cohesive research groups, with Faisal Anwer, R. Imam, and S. Afzal emerging as central and recurrent contributors across multiple studies. These authors appear as high-degree nodes, indicating frequent collaborations and a pivotal role in connecting disparate research clusters.

F3. Figure 19 represents the keyword co-occurrence network graph offers a comprehensive visualization of the thematic structure and research focus areas within lightweight cryptographic solutions for IoT security. The multicolor nodes represent distinct clusters of semantically related keywords, while the colored edges

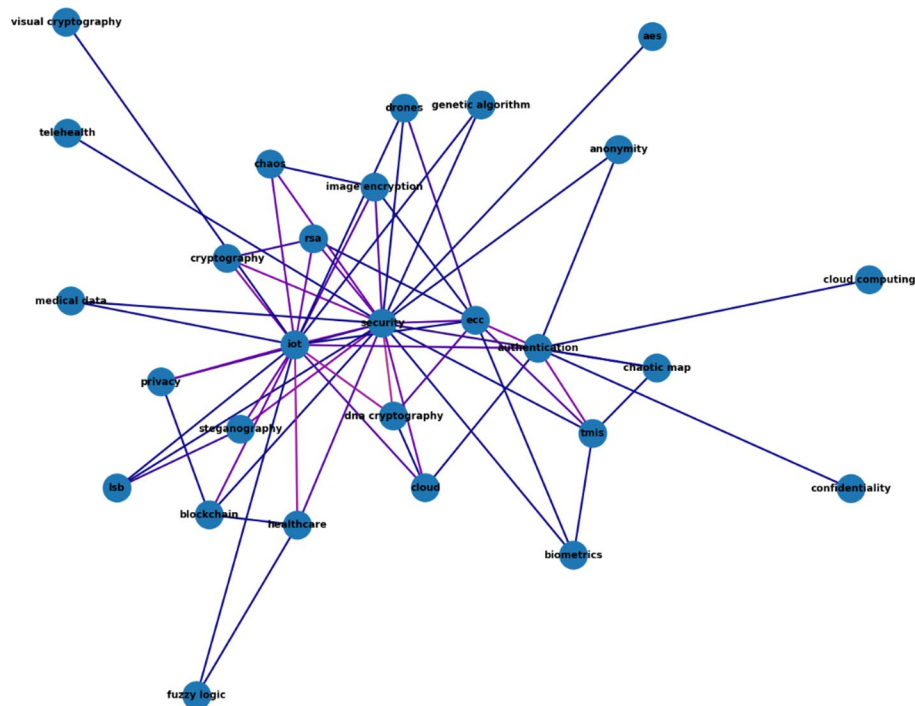


Fig. 19 Keyword co-occurrence network for primary studies

reflect the strength of co-occurrence between keyword pairs across primary studies. Prominent and frequently co-occurring terms such as “IoT,” “security,” “cryptocurrency,” “ECC,” and “lightweight” form dense cores within the graph, indicating their central role in ongoing research. The emergence of thematic clusters around keywords like “blockchain,” “fuzzy logic,” “stream cipher,” and “authentication” further suggests diversification in applied techniques and growing interdisciplinary integration.

- F4. Among the reviewed studies, 84.4% address confidentiality, 64.9% focus on authentication, and 98.7% on integrity. However, 78% tackle all three security objectives simultaneously, revealing a gap in holistic cryptographic frameworks.
- F5. Asymmetric cryptography, especially Elliptic Curve Cryptography (ECC), was employed in over 30% of the studies for its computational efficiency and key size benefits. Symmetric methods were used in about 10.4%, often in applications requiring speed and low overhead.
- F6. Nearly 29.9% of studies proposed hybrid models (e.g., AES + ECC, chaos + ECC), which combine the speed of symmetric ciphers with the security of asymmetric key exchange mechanisms, offering balanced trade-offs for IoT devices.
- F7. 25.9% of the studies incorporated artificial intelligence (fuzzy logic, SVM) or blockchain for intrusion detection, authentication, and secure data logging, indicating a growing convergence of lightweight cryptography with intelligent and decentralized systems.
- F8. Only 10.4% of the papers explored the integration of steganography with cryptography for covert communication, highlighting an under-explored area despite its potential in enhancing confidentiality in IoT.

- F9. ECC remained the most frequently used algorithm, followed by RSA, AES, and emerging chaotic systems like Logistic, Lorenz, and Tent Maps, particularly in stream cipher design and key generation schemes.
- F10. Application-domain focus reveals healthcare IoT as the most extensively studied environment, particularly due to the sensitivity of medical data and demand for strong authentication. Other domains such as smart homes, industrial IoT, and environmental monitoring appear less frequently, indicating opportunities for expanding lightweight cryptographic applications beyond healthcare.
- F11. Many proposed models included Shannon entropy and NIST SP 800–22 test results, showing improved unpredictability in keystreams and ciphertexts, crucial for resisting statistical and differential attacks.
- F12. Studies benchmarked encryption time, memory usage, energy consumption, and throughput. In many cases, proposed models outperformed conventional algorithms like RSA and AES when tested in lightweight environments.
- F13. While many proposed schemes report promising results, practical validation remains limited: only ~12% of studies (9 out of 77) tested implementations on physical IoT devices such as Raspberry Pi, Arduino, or NodeMCU. The remaining ~88% relied solely on simulations or theoretical analysis, highlighting a gap between experimental validation and real-world deployment.
- F14. Major gaps include the lack of adaptive cryptographic frameworks, underexplored optimization-based key management, and insufficient standardized benchmarking across studies, hindering the reproducibility and scalability of many proposed solutions.
- F15. The symmetric algorithms (AES, ChaCha20) excel in scalability and latency but depend on strong key distribution for security. ECC provides robust security with acceptable latency and better scalability than RSA, making it practical for IoT authentication. RSA, while secure, is too resource-intensive for constrained environments. Hybrid schemes achieve the best balance, combining the strong authentication of ECC with the low-latency performance of AES or ChaCha20, making them the most viable for real-world IoT deployments.
- F16. Several gaps persist, including the absence of adaptive cryptographic frameworks, limited benchmarking on physical IoT platforms, and minimal consideration of post-quantum readiness. To address these gaps, future research should prioritize the development of adaptive schemes that dynamically adjust cryptographic strength based on device resources, conduct systematic hardware evaluations to validate scalability, and investigate lightweight post-quantum algorithms (e.g., lattice-based, hash-based signatures) for IoT resilience against emerging threats.

6.1 Lightweight cryptography in multi-layer IoT architectures

Lightweight cryptography plays a distinct role at each layer of the IoT architecture, and its effectiveness must be evaluated in a multi-layer context rather than in isolation (see Fig. 20). At the perception layer, symmetric algorithms such as AES-128, ChaCha20, and chaos-based ciphers ensure real-time protection of sensor data with low latency and minimal resource usage, though they require efficient key distribution. At the network layer, ECC-based mutual authentication and blockchain-assisted trust mechanisms secure communications between devices, gateways, and fog nodes while reducing

Lightweight Cryptography in Multi-layer IoT Architectures

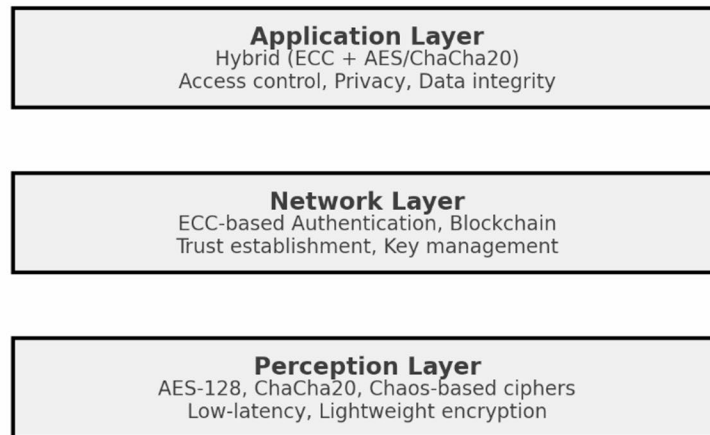


Fig. 20 Multi layer IoT cryptography

dependence on centralized authorities. At the application layer, hybrid approaches that combine ECC with AES or ChaCha20 provide scalable key management, data confidentiality, and integrity for domains such as healthcare, smart homes, and industrial IoT. This layered perspective highlights that no single algorithm is sufficient for end-to-end IoT security; instead, schemes must be tailored and integrated across layers to balance security, scalability, and latency in real-world deployments.

7 Limitations of the survey

Despite the comprehensive scope and methodological rigor of this systematic review, several limitations exist that merit acknowledgment. Firstly, although the study employed a multi-phase filtration process and adhered to the systematic review guidelines established by Kitchenham and Charters, the final inclusion of 77 primary studies may not represent the full breadth of global research in lightweight cryptography for IoT. Language restrictions to English-only publications and reliance on indexed databases such as IEEE, Elsevier, Springer, and ACM may have led to the exclusion of valuable non-English or regionally published research. Additionally, keyword-based search strategies despite being carefully constructed using synonym expansion and Boolean logic—are inherently limited by the variability in author terminology. For instance, studies employing lightweight cryptographic techniques may not explicitly mention terms such as “lightweight,” “IoT,” or “encryption” in their titles, abstracts, or keywords, leading to their inadvertent omission.

Secondly, while the bibliometric analyses of keyword co-occurrence and author collaborations offer valuable insights into research trends and intellectual structures, they are limited by metadata granularity. Factors such as institutional affiliations, funding sources, geographic origin, or interdisciplinary collaboration dynamics are not fully captured, which could otherwise enrich the interpretation of research influence and maturity within specific regions or institutions.

Thirdly, although the review categorizes algorithms by type (e.g., symmetric, asymmetric, hybrid), the assessment of performance such as energy efficiency, computational complexity, and throughput is often drawn from simulation studies rather than

real-world implementation. Most reviewed models lack experimental validation on embedded IoT hardware platforms like Raspberry Pi, Arduino, or ESP32. Consequently, their claimed lightweight nature may not hold under practical constraints involving fluctuating network loads, memory bottlenecks, or real-time latency. Similarly, very few studies offer comprehensive benchmarks using standardized testing suites such as NIST SP 800–22 and compare cryptographic strength via metrics like Shannon entropy, avalanche effect, and correlation coefficients in consistent environments.

Furthermore, although the review highlights the integration of emerging paradigms such as blockchain, steganography, chaotic systems, and artificial intelligence with cryptography, most of these integrations remain at a conceptual or prototype stage. These hybrid approaches, while innovative, often lack holistic evaluation frameworks and security proofs, and are not tested for interoperability or scalability in heterogeneous IoT ecosystems. For instance, blockchain-augmented lightweight protocols may introduce latency due to consensus mechanisms, and AI-driven adaptive cryptography may require computational resources exceeding those available on constrained IoT devices.

Interoperability remains a significant challenge for deploying lightweight cryptography in heterogeneous IoT systems. Devices from different vendors often vary in processing power, memory, and supported protocols, which leads to inconsistent performance and fragmented security. The absence of standardized lightweight cryptographic frameworks further complicates seamless integration across smart homes, healthcare, and industrial IoT. Addressing these issues requires protocol standardization and certificate-less authentication schemes to ensure scalable, multi-vendor compatibility.

Lastly, this review does not cover post-quantum cryptographic approaches, which are gaining momentum as future-proof security solutions, particularly for long-lived IoT deployments. The rapidly evolving nature of cryptographic standards and the emergence of quantum-resistant algorithms, federated learning in cryptographic key management, and decentralized identity protocols suggest that ongoing updates to this review will be essential. These limitations underscore the need for further empirical studies, hardware-level benchmarking, and the development of unified evaluation frameworks that can support the reproducibility, scalability, and standardization of lightweight cryptographic solutions in IoT.

7.1 Limitations in real-world smart home deployments

In real-world smart home environments, the deployment of lightweight cryptography faces several challenges. First, device heterogeneity means that protocols optimized for one class of devices may not scale effectively across the wide variety of sensors, actuators, and appliances typically found in smart homes. Second, although lightweight algorithms reduce computational overhead, they may still stress ultra-constrained nodes, leading to trade-offs between security strength and device longevity. Third, dynamic key management in dense smart home networks introduces scalability and communication complexities. Interoperability is another concern, as devices from multiple vendors often lack standardized lightweight cryptographic support, weakening system-wide security. Finally, usability and physical security issues remain critical: frequent re-keying or updates can hinder user experience, and resource-limited devices remain vulnerable to side-channel and physical attacks. These limitations underscore the need for

standardized, adaptive, and user-centric lightweight cryptographic frameworks in future smart home deployments.

8 Future direction

Future research on lightweight cryptography for IoT should focus on designing energy-adaptive protocols that dynamically adjust key lengths, encryption rounds, or cipher selection based on device energy levels and workload, ensuring sustainability in battery-powered systems. To validate practicality, hardware-aware benchmarking on platforms such as NodeMCU, Raspberry Pi, and ARM Cortex-M is essential, providing empirical data on execution time, memory footprint, energy use, and communication costs under realistic IoT workloads. Additionally, efforts must prioritize standardization and interoperability to enable seamless operation across heterogeneous devices, alongside a user-centric design approach that minimizes latency, reduces user intervention, and maintains transparent security. Finally, integrating lightweight cryptography with emerging technologies such as AI-driven key management, blockchain-based trust models, and privacy-preserving methods like fully homomorphic encryption and differential privacy can yield scalable, resilient, and future-proof security frameworks.

Future research should focus on the following actionable directions:

- *Energy-Adaptive Cryptography* – Develop schemes that adjust encryption strength and key size based on device energy levels and workload.
- *Hardware-Aware Evaluation* – Benchmark lightweight schemes on real IoT hardware (e.g., NodeMCU, Raspberry Pi, ARM Cortex-M) for realistic performance insights.
- *Standardization & Interoperability* – Create lightweight protocol standards to ensure seamless operation across heterogeneous IoT devices and vendors.
- *User-Centric Design* – Prioritize usability by minimizing latency, reducing user intervention, and ensuring transparent security operations.
- *Emerging Tech Integration* – Combine lightweight cryptography with AI, blockchain, and privacy-preserving methods (e.g., FHE, differential privacy) to strengthen resilience.

Future research should also consider the impact of lightweight cryptography on usability and user experience in IoT environments. While current schemes succeed in reducing computational and communication costs, they may still introduce latency, frequent re-keying, or interoperability issues that affect seamless operation in domains such as smart homes and healthcare. Addressing these factors requires the development of adaptive, user-centric protocols that balance strong security with minimal disruption to device performance and daily use. Exploring this intersection of security and usability can ensure broader acceptance and sustainable deployment of lightweight cryptographic frameworks.

8.1 Lightweight cryptography in the context of post-quantum threats

The emergence of quantum computing poses significant risks to traditional lightweight cryptographic schemes, particularly those relying on asymmetric primitives such as ECC and RSA, which are vulnerable to Shor's algorithm. Symmetric algorithms like AES and ChaCha20 remain more resilient, but require longer key sizes (e.g., AES-256) to maintain equivalent post-quantum security, which can increase computational overhead for

constrained IoT devices. Recent advances in post-quantum cryptography (PQC), especially lattice-based schemes such as Kyber and NTRU, and code-based approaches like BIKE, are being investigated for IoT, though their larger key sizes and higher memory requirements present challenges for resource-limited nodes. A promising path forward is hybrid cryptography, where lightweight schemes (AES/ChaCha20 for data encryption, ECC for short-term key exchange) are combined with PQC primitives for long-term resilience. Future work should benchmark these hybrid models on IoT platforms (e.g., Raspberry Pi, NodeMCU, ARM Cortex-M) to assess their feasibility in balancing efficiency, scalability, and post-quantum security guarantees.

9 Conclusion

In addressing the core research questions namely, how lightweight cryptographic models ensure confidentiality, integrity, and authentication under stringent IoT resource constraints, how they compare in computational efficiency, how ECC can be optimized, and how emerging technologies such as AI, blockchain, and steganography integrate into these solutions this review provides a comprehensive synthesis of the field. By systematically analyzing 77 primary studies published between 2006 and 2025, we classified and compared lightweight block and stream ciphers, ECC-based authentication frameworks, and hybrid cryptographic schemes across diverse IoT environments.

Our findings highlight that performance metrics such as execution time, memory footprint, energy consumption, and statistical security measures (e.g., entropy, avalanche effect, correlation analysis, NIST randomness tests) are critical in assessing real-world applicability. The review demonstrates that ECC-based protocols dominate due to their balance of efficiency and robustness, while hybrid approaches with AI, blockchain, and steganography are emerging as next-generation solutions.

Despite these advances and thematic insights, the review reveals enduring limitations that must be addressed to translate theoretical cryptographic designs into deployable, scalable IoT solutions. These include limited hardware validation, lack of adaptive models, and insufficient exploration of post-quantum readiness. Future research must therefore bridge this gap by integrating lightweight cryptography with adaptive, AI-driven intelligence, standardized benchmarking, and regulatory compliance (GDPR, HIPAA).

Emerging trends identified in the review suggest a strong shift toward designing adaptive, cross-layer security architectures capable of dynamically adjusting cryptographic complexity based on device constraints, application context, and real-time threat levels. Moreover, the need to incorporate quantum-resilient primitives, such as lattice-based or code-based encryption, is becoming increasingly urgent as IoT systems evolve toward long-lived deployments in critical infrastructure and national security domains. This study provides a foundational knowledge base and a strategic roadmap for researchers, developers, and system architects aiming to design next-generation cryptographic solutions that are secure, scalable, energy-efficient, and future-proofed for the evolving threat landscape in IoT Table 13.

Table 13 Open challenges and future research directions in lightweight cryptography for IoT

Open challenge	Research gap	Future research direction
Scalability in Large-Scale IoT Networks	Many lightweight schemes are optimized for small networks but lack evaluation in large, heterogeneous IoT deployments.	Develop adaptive cryptographic frameworks with dynamic key management and hierarchical trust models suitable for large-scale IoT ecosystems.
Energy and Resource Constraints	High security often increases computational/energy overhead, limiting deployment on ultra-constrained devices.	Design ultra-lightweight ciphers and hardware-accelerated cryptographic modules for wearables, sensors, and embedded IoT nodes.
Post-Quantum Security	Current lightweight ciphers may be vulnerable to quantum attacks; few works address PQC in IoT.	Explore post-quantum lightweight cryptography (e.g., lattice-based, code-based) tailored for constrained IoT environments.
Integration of AI and Cryptography	Limited studies use AI/ML for adaptive security or key generation.	Employ AI-driven key generation, anomaly detection, and adaptive cryptography for context-aware IoT security.
Steganography and Hybrid Approaches	Very few works combine steganography with lightweight cryptography.	Investigate hybrid cryptography + steganography for dual-layer security, especially in medical and drone-based IoT systems.
Standardization and Benchmarking	Lack of unified benchmarks; results vary across platforms and datasets.	Establish standard IoT testbeds and benchmark datasets for evaluating lightweight cryptographic solutions.
Interoperability Across IoT Layers	Schemes often target a single IoT layer (device, network, cloud) without cross-layer optimization.	Design cross-layer lightweight protocols ensuring end-to-end confidentiality, authentication, and integrity.
Privacy-Preserving Data Sharing	Limited focus on secure multi-party computation and federated models in IoT.	Explore federated learning + lightweight cryptography and homomorphic encryption for privacy-preserving IoT analytics.

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Shaharyar Alam Ansari and Salman Ali jointly conceptualized and designed the systematic review framework. Shaharyar Alam Ansari conducted the literature search, data extraction, and performed the comparative analysis of cryptographic schemes. Salman Ali contributed to drafting and refining sections on authentication, confidentiality, and ECC-based protocols. Shaharyar Alam Ansari prepared the figures and tables. Both authors contributed to interpreting the results, identifying research gaps, and reviewing the final manuscript.

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